

The Foreign Policy Impact of Trade-Based Status Gains: When China Overtakes the US as Top Trade Partner

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Abstract

How do trade-based status gains affect foreign policy alignment? Existing IR research emphasizes status anxiety and status seeking, but pays little attention to upward status shifts. We argue that rank reversals in trade hierarchies, especially when a rising power displaces the hegemon, create a salient status shock that can produce foreign policy realignment beyond gradual trade growth. We test this claim in Brazil, where China overtook the United States as top trade partner in 2009, using Synthetic Difference-in-Differences on UNGA ideal-point distance. The estimate implies a statistically significant 41% reduction in Brazil's ideological distance to China after the reversal ($p = 0.032$). To probe mechanism implications, we analyze Brazil's main newspaper with NLP and find evidence consistent with a post-2009 increase in China-related salience, particularly trade-focused coverage. We then examine cross-country cases in which China displaced the United States as top partner. Because four of the eight treated countries experience treatment reversals (the USA regains the top position), we employ the counterfactual estimator of Liu, Wang, and Xu (2024) with interactive fixed effects, which accommodates non-absorbing (switching) treatment. The cross-country estimate confirms a similar pattern. Exit-aligned analysis further shows that the alignment effect dissipates when the USA regains the top position, providing direct evidence for a salience-driven mechanism. These results suggest that discrete status milestones, not only continuous trade deepening, can reshape foreign policy behavior. The findings challenge claims that status gains are mostly illusory and motivate research on status effects in trade, investment, aid, and other economic ties.

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1 Introduction

“China overtakes US as Brazil’s top trade partner”. That was a headline in many Brazilian newspapers in 2009 (Rodrigues 2009). The BBC ran a similar headline in 2021, when China surpassed the United States as the EU’s biggest trading partner (BBC News 2021). Sensing the symbolic weight of the milestone in Brazil, President Lula da Silva promptly announced a high-profile trip to Beijing, flanked by an unusually large business delegation.

Do discrete rank reversals in trade hierarchies — particularly when a rising power displaces the hegemon — produce foreign policy realignment beyond what continuous trade growth predicts?

The status literature (Renshon 2016; Ward 2017; Dafoe, Renshon, and Huth 2014) focuses almost exclusively on status deficits and conflict. It offers limited theoretical guidance predicting that a state gaining trade-based status would receive diplomatic concessions. The literature on status effects of rank changes exists in domestic politics (Anagol and Fujiwara 2016; Fujiwara and Sanz 2020) but has never been extended to international politics.

An extensive empirical literature shows that as a country’s trade dependence on China accumulates, its UN-voting distance from Beijing tends to fall (Kastner and Pearson 2021; Yan and Zhou 2021; Morgan 2019; Flores-Macías and Kreps 2013), the same being true for Brazil and China in particular (Neto and Malamud 2015; Mourón and Urdinez 2014). Analysts interpret this association in three ways: coercive leverage, i.e., China rewards or punishes to shape votes (Tanner 2007; Reilly 2013), interest redefinition (growing commerce makes cooperation intrinsically valuable), and societal pressure (business lobbies or public opinion pull leaders toward Beijing). What these perspectives share is the assumption that influence scales smoothly with commercial exposure: the more trade, the more alignment — no particular trade share or moment is distinguished from another.

Research in cognitive psychology and political communication, by contrast, emphasizes that actors respond disproportionately to salient thresholds – events that break routine patterns and become focal in media and politicians’ agendas (Baekgaard et al. 2019; Sheffer et al. 2018; Edwards and Wood 1999; Mintz and Geva 1997; Mercer 2017).

While most studies of status emphasized how lacking status drives risk-acceptant behavior and conflict, there is only anecdotal evidence that status yields tangible diplomatic dividends (Götz 2021). As a result, we know surprisingly little about the downstream effects when a state’s status actually rises.

We argue that rank reversals deliver discrete status changes that, processed as coarse categories, trigger disproportionate media salience and foreign policy realignment, beyond what gradual trade accumulation predicts. This effect is more likely when other cues increase salience, such as the former trade partner being the hegemon or the reversal overcomes a long-run historic pattern.

To develop and initially probe our theory, we focus on the most likely case: Brazil. The US was Brazil’s top trade partner for approximately 80 years, making the displacement maximally salient. Brazil has a free press (media channels can operate freely); Brazil is a regional power, not a small state, which makes coercion unlikely. Most-likely cases are well-suited for theory generation with confirmatory value: if the effect doesn’t appear here, it’s unlikely to appear anywhere. We follow a structured hybrid approach: the reduced-form prediction (rank reversal → alignment) is tested in Brazil (SDiD) and cross-country (DiD), while the salience mechanism is proposed and probed using the Brazilian case and NLP media evidence.

To credibly show that the overtaking of the US by China as Brazil’s top trade partner has a causal effect on Brazil’s foreign policy orientation, we use a synthetic difference-in-differences approach. We measure the outcome as the absolute distance in UNGA ideal point estimation (Bailey, Strezhnev, and Voeten 2017) between Brazil and China over 1997-2015. The SDiD point estimate corresponds to an estimated 40.8% reduction in Brazil’s ideological distance from China in UNGA votes.

As Brazil is a single case, we test the theory that status change impacts foreign policy using a cross-country design on all countries where the United States was the top trade partner. We compare those where China eventually displaced the US (the treatment) with those where it did not (control group), using a counterfactual estimator with interactive fixed effects (Liu, Wang, and Xu 2024) that accommodates treatment switching – four of the eight treated countries see the USA regain the top position. The results are consistent with the case study, and exit-aligned analysis shows the effect dissipates when treatment turns off.

To illuminate the mechanisms behind the effect of status change, we perform a deep analysis of the Brazilian case, showing a plausible channel through which rank appears as a cue that may trigger change in policy making: the media’s role in publicizing and thus legitimizing this newfound status.

This study provides the first causal evidence that trade rank reversals produce foreign policy realignment, with suggestive evidence that a media-salience mechanism contributes to this effect. It questions the idea in the literature that status gains are mostly illusory (Mercer 2017). It also has broad implications for the study of economic ties with an emerging power and the foreign policy of countries. Similar status change effects may also shape how other economic variables (such as Foreign Direct Investment, aid, and loans) impact the foreign policy of countries.

2 Trade, Status, and Political Alignment

An extensive literature examines how trade with China affects foreign policy. The proposed mechanisms vary – coercive leverage, where China rewards or punishes to shape votes (Tanner 2007; Reilly 2013); interest redefinition, where deepening economic ties create vested interests that lobby on behalf of Beijing (Kastner and Pearson 2021); soft power and societal pressure, where economic integration transforms public and elite opinion (Morgan 2019; Kastner and Pearson 2021); and structural power, where China’s rise as a great power pulls countries toward its orbit (Struver 2014; Schenoni and Leiva 2021; Neto 2011; Mourón and Urdinez 2014). What these perspectives share is the assumption that influence scales smoothly with commercial exposure: the more trade, the more alignment – no particular trade share or moment is distinguished from any other.

Yet the empirical record shows reality is messier. Reverse causality remains a concern: political alignment may drive trade rather than the reverse (Haibin 2010; Coutinho and Rodriguez 2024; Fuchs and Klann 2013; Yan and Zhou 2021; Chen and Zhou 2021; Aiyar, Malacrino, and Presbitero 2024)¹. More substantively, if gradual trade growth consistently produced alignment, we should observe a monotonic relationship. Instead, Carmody, Dasandi, and Mikhaylov (2020) find

¹The most ambitious causal study in this tradition, Flores-Macías and Kreps (2013), employs fixed-effects with lagged dependent variables and two-stage least squares. Both strategies face well-documented identification problems: the lagged dependent variable violates the no-carryover assumption required for FE consistency (Blackwell and Glynn 2018; Imai and Kim 2019, 2021), and the instrument (lagged energy production) likely fails the exclusion restriction (Mellon 2021; Bastardoz et al. 2023; Lal et al. 2024).

the opposite for Africa: countries that increased trade with China voted *more* with the United States, not less – consistent with hedging rather than bandwagoning. Garcia-Herrero, Storella, and Weil (2024) show that UNGA alignment with China has remained remarkably stable despite China’s dramatic economic rise, suggesting that continuous trade growth alone does not move the needle on foreign policy. If gradual trade growth does not consistently produce alignment, what does?

2.1 Why Rank Reversals Matter

Status has been extensively studied in international relations (Dafoe, Renshon, and Huth 2014; Miller et al. 2015; Renshon 2016; Renshon, Dafoe, and Huth 2018; Duque 2018; Parlar Dal 2019; Götz 2021; Røren 2024). A standard definition is the ranking or relative standing of a state (MacDonald and Parent 2021). This definition, however, conflates a continuous measure (relative standing) with an ordinal one (rank position). Being first is qualitatively different from being second, and that distinction matters for how actors process information and allocate attention. We distinguish between *trade status* – the ordinal position of a trade partner in a country’s hierarchy of commercial relationships – and *status shock* – the discrete event of a rank reversal. In an experimental setup, status is the treatment; rank reversal is the manipulation. Our argument concerns the downstream consequences of status shocks, not status seeking, status anxiety, or military conflict – the phenomena that dominate the existing IR status literature.

The micro-foundation for why rank matters comes from coarse categorization – the tendency to rely on broad, easily processed bins rather than fine-grained metrics when confronting complexity (Graeber, Roth, and Sammon 2025; Enke 2024). Actors – media editors, policymakers, business leaders – attend to categories like “top trade partner” as well as tracking continuous trade shares. The relevant categories here are “US = top partner” (a cognitive default entrenched by decades of commercial primacy) versus “China = top partner” (a disruptive reclassification). The rank reversal does not bring new information about trade volumes – those data are continuously available. What it brings is a *categorical reclassification* that captures attention precisely because it disrupts an entrenched bin. Even redundant information shifts attention when it alters salient categories (Conlon 2025; Bordalo, Gennaioli, and Shleifer 2022), and people tend to neglect the correlated structure of information and treat novel framings as independent signals (Enke and Zimmermann 2017).

The effect of rank on political decisions is well documented in domestic politics. Rank reversals produce bandwagon effects in elections (Anagol and Fujiwara 2016; Granzier, Pons, and Tricaud 2023), regression discontinuities at the boundary of winning versus losing affect subsequent political careers (Folke, Persson, and Rickne 2016; Fujiwara and Sanz 2020), and similar rank-based thresholds coordinate behavior across diverse actors. The rank reversal functions as a focusing event by breaking an entrenched cognitive category. Yet no study has examined whether rank effects operate in international politics, despite the plausibility that they do. Status is widely acknowledged to matter in world politics, but there is no quantitative study estimating how a positive status shock translates into concrete policy changes or political alignment.

2.2 Two Claims: Attention and Policy Response

Our theory rests on two distinct claims. *Claim 1 (Attention)*: A rank reversal produces a disproportionate increase in media salience. The micro-foundation is coarse categorization: the reclassification from “second partner” to “first partner” triggers a categorical reframing in news coverage. The amplification occurs through agenda-setting (Edwards and Wood 1999) – the media transforms the categorical reclassification into a focal event for national attention. Since this is a bounded rationality reaction, it is plausible that it would be temporary: as the new status becomes the cognitive default, salience fades. We provide evidence for this claim directly using NLP analysis of Brazilian newspaper coverage.

Claim 2 (Policy Response): The salience shock translates into foreign policy realignment through complementary channels. First, a cognitive channel: poliheuristic theory (Mintz 2004, 2005; Mintz and Geva 1997) predicts that decision-makers use non-exhaustive search heuristics, and the salience spike causes the new trade status to enter their consideration sets, even if the underlying economic relationship was previously discounted. The role of cues and heuristics in foreign policy decisions has been discussed in international relations (Jabarian and Sartori 2024). Second, an audience cost channel: once the media publicizes the new status, leaders face reputational incentives to align discourse and action with the newly salient relationship (Fearon 1994; Tomz 2007). Third, a coordination channel: the rank reversal creates a focal point in the Schelling sense around which business lobbies, diplomats, and public opinion coordinate their demands for policy adjustment.

These channels are complementary, and our design does not distinguish among them. The SDiD identifies their combined reduced-form effect.

We are explicit about the limits of our design. The SDiD estimates the combined reduced-form effect of the rank reversal on alignment through all channels. The NLP media analysis provides suggestive evidence for Claim 1 (attention). Claim 2 – how salience translates into votes – is theorized but not causally identified. We provide convergent evidence consistent with an attention channel, while acknowledging that the policy response channel remains theorized rather than directly identified.

2.3 Hypotheses

From the theoretical building blocks above, we derive four testable hypotheses.

The first hypothesis is the main effect of interest and is related to the reduced-form specification.

Hypothesis 1 (Main Effect): Rank reversals – where China displaces the incumbent top trade partner – produce a discrete reduction in UNGA voting distance from China, beyond what gradual trade growth predicts. This follows from coarse categorization (the rank reversal is a category disruption) that opens a window for policy change.

Our theorization also makes predictions about mechanisms, of which we specify here the ones we can test or provide evidence consistent with in the paper.

Hypothesis 2 (Hegemon Moderation): The effect is larger when an emerging power displaces the hegemon, compared to when the displaced partner is any other country. Coarse categorization theory predicts that disrupting a more entrenched cognitive category, and “US = top partner” is among the most deeply entrenched, produces a larger attention shock (Graeber, Roth, and Sammon 2025).

Hypothesis 3 (Media Salience): Media coverage of the trade relationship increases disproportionately at the moment of the rank reversal, rather than tracking gradual trade growth. This is the direct observable implication of Claim 1 and is testable with the NLP analysis.

Hypothesis 4 (Temporal Attenuation): The alignment effect attenuates over time as the new status becomes the cognitive default. Salience is by nature temporary – the new rank normalizes and media attention shifts to other topics. This hypothesis is particularly discriminating: structural

bandwagoning, coercion, and interest-group theories all predict *persistent* effects. Only a salience mechanism uniquely predicts transience.

2.4 Scope Conditions

The status-salience mechanism operates under conditions that follow from the theoretical building blocks above. First, the displaced partner must be geopolitically significant, ideally the hegemon. Coarse categorization theory predicts that disrupting a more prominent cognitive category produces a larger attention shock. Second, the media amplification channel requires a free or active press; in countries with restricted press freedom, the rank reversal may go unreported, muting the salience mechanism. This follows directly from the agenda-setting literature (Edwards and Wood 1999). Third, the duration of the prior relationship matters: the longer the displaced partner held the top position, the more entrenched the existing category and the more disruptive the reversal (Mintz and Geva 1997). Fourth, democratic accountability moderates the effect, as media salience is more consequential where public opinion constrains foreign policy choices. Fifth, the rank reversal should be exogenous to deliberate trade diversification policy – if a government actively pursued the rank reversal as part of a broader realignment strategy, the effect captures policy intent rather than the salience mechanism. In the cross-country analysis, we test scope condition 1 directly by comparing specifications where China displaced the United States versus any partner, and scope conditions 2 and 3 enter as covariates (press freedom and US streak).

2.5 Causal Framework

The causal claim of the present paper can be summarized by the simplified Directed Acyclic Graph (DAG) – a diagram that visually represents assumed causal relationships – below. We represent the main variables and proposed mechanisms, without any confounding variables, to clarify the argument. We discuss the identification strategy in the next section.

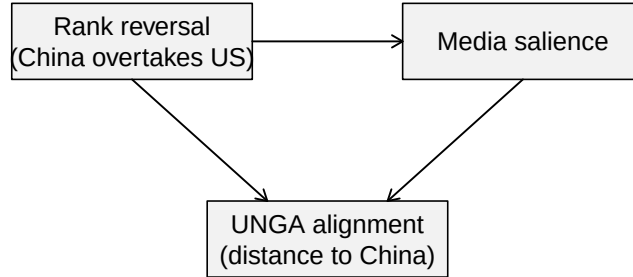


Figure 1: Simplified DAG of the causal relationship between China’s status as top trade partner and foreign policy alignment. Note: Unit = conceptual diagram (no numeric scale). Standard error = not applicable. Cluster = not applicable. Time window = conceptual graph; treatment onset is 2009 in Brazil and country-specific in panel analyses. Treatment definition = Treatment indicator equals 1 in any year China is currently the top export destination; turns off if USA regains the position (switching treatment).

3 Methodology

3.1 Brazilian Case

For the Brazilian case, we employ the synthetic difference-in-differences (SDiD) method (Arkhangelsky et al. 2021), which combines the strengths of standard DiD and the synthetic control method (SCM). SDiD assigns unit weights so that the synthetic control matches the treated unit’s pre-treatment trajectory, and time weights so that the comparison focuses on periods closest to treatment onset. Unlike SCM, it retains unit fixed effects, accounting for time-invariant unobservables. Unlike standard DiD, it does not require that all control units follow the same trend as the treated unit – an assumption that is implausible here given the 2008 economic crisis and other concurrent shocks. A formal comparison of the DiD, SCM, and SDiD estimating equations is provided in the Appendix.

Following Hirshberg and Klosin (2024), we augment the model with time-varying covariates X_{it} for more accurate pre-treatment matching. The estimating equation is:

$$(\hat{\tau}_{ATT}^{\text{SDiD}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}, \hat{\gamma}) = \arg \min_{\tau_{ATT}, \mu, \beta, \alpha, \gamma} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \mu - \alpha_i - \beta_t - D_{it} \tau_{ATT} - X_{it} \gamma)^2 \hat{w}^{\text{SDiD}} \hat{\lambda}_t^{\text{SDiD}}$$

where $\hat{w}^{\text{SDiD}}$ are unit weights and $\hat{\lambda}_t^{\text{SDiD}}$ are time weights, both estimated from pre-treatment data. Standard errors are computed using the placebo variance estimator proposed by Arkhangelsky et al. (2021), which estimates the variance of the treatment effect by computing pseudo-treatment effects on each control unit.

3.2 Cross-Country Design

Although the Brazilian case is a credible strategy to study the causal effect of status change, it is still $n = 1$, and does not allow generalization beyond its specific features. Furthermore, we cannot rule out that the explanation resides in something specific to Lula da Silva’s or the Workers’ Party’s presidency in the period considered, or rule out alternative explanations of potential confounders that happened at a similar time, such as the global economic crisis in the aftermath of Lehman Brothers’ fallout and how the 2008 Beijing Summer Olympics changed China’s status.

To address such challenges, we examine all countries in our sample that had the US as their top trade partner and consider the effect of China overtaking the US. An important empirical feature is that this treatment is not absorbing: four of the eight treated countries experience treatment reversals – the USA regains the top position after China held it temporarily. Standard staggered DiD estimators such as Callaway and Sant’Anna (2021) assume absorbing treatment and cannot accommodate this switching pattern.

We therefore adopt the counterfactual estimator with interactive fixed effects (IFE) proposed by Liu, Wang, and Xu (2024) as our main cross-country specification. The IFE framework estimates counterfactual outcomes for treated observations using untreated observations and pre-treatment data from treated units. The IFE model relaxes strict parallel trends by allowing for latent common

factors with heterogeneous loadings, and cross-validation selects the optimal number of factors r^* . Crucially, `fect` natively handles treatment switching: it estimates entry effects (what happens when treatment turns on) and exit effects (what happens when treatment turns off), providing a direct test of whether the alignment shift persists or dissipates after the status shock fades.

To further investigate entry and exit dynamics, we complement `fect` with `PanelMatch` (Imai, Kim, and Wang 2023), a matching-based estimator for time-series cross-sectional data that estimates both the Average Treatment Effect on the Treated (ATT, effect of switching on) and the Average Treatment Reversal Effect on the Treated (ART, effect of switching off). As a robustness check, we also report results from the Callaway and Sant’Anna (2021) estimator restricted to absorbing-only countries (Appendix).

3.3 Identification Assumptions and Diagnostics

For the Brazilian SDiD design, identification requires that no unobserved shocks differentially affecting Brazil occur at treatment timing after weighting. This assumption is not fully testable. We therefore treat pre-treatment fit and in-time placebo exercises as partial diagnostics that are consistent with identification, rather than definitive tests.

For the cross-country IFE design, identification requires strict exogeneity: treatment assignment must be independent of idiosyncratic outcome shocks, conditional on the factor structure (Liu, Wang, and Xu 2024; Bai 2009). This is more flexible than parallel trends (it allows for heterogeneous trends via latent factors) but more restrictive regarding feedback from outcome shocks to treatment timing. We assess model adequacy via two diagnostic tests recommended by Liu, Wang, and Xu (2024). First, a placebo test on pre-treatment periods: under correct specification, estimated effects in pre-treatment periods should be negligibly small. Because statistical power is limited with few treated units, we employ the equivalence approach rather than the conventional test – the equivalence test provides positive evidence *for* the absence of pre-trends, whereas the conventional test can only fail to reject their presence (Liu, Wang, and Xu 2024; Chiu et al. 2025). Second, a carry-over test assesses whether treatment effects linger after exit, using the same equivalence framework.

Accordingly, mechanism analyses are interpreted as suggestive evidence, not as causally identified mechanisms.

3.4 Data and Variables

To fit the model, current implementations of SDiD require a complete panel. Thus, we assemble a country-year panel that merges several datasets, covering 96 countries over 19 years (1997 - 2015).

Our outcome variable, vote similarity between a country and China at the United Nations General Assembly (UNGA), is based on ideal point estimates derived from UNGA vote data from Bailey, Strezhnev, and Voeten (2017). We use the median estimate of the ideal points of each country per year and compute the absolute distance between China in a given year and each other country-year in the database. The lower the score, the closer the foreign policy between the countries. It measures state preferences apart from the agenda, which is important whenever a comparison is made over time, as is the case here. It is a better measure over time than raw similarity indices, such as the percentage of similar votes in a year (Neto and Malamud 2015), or any other measures, such as the S-score (Signorino and Ritter 1999).

UNGA vote affinity has been used as a measure of shared state preference in the literature due to its comprehensive membership coverage and the regular availability of data (Potrafke 2009; Strüver 2016). The topics covered by UNGA resolutions primarily focus on political dimensions, particularly humanitarian and social issues, with a small fraction of resolutions addressing economic and financial matters (Strüver 2016).

For the Brazilian SDiD, the treatment indicator D_{it} equals 1 in the first year that China became Brazil's leading trade partner, in 2009, and in all subsequent years. For the cross-country analysis, D_{it} equals 1 in any year China is currently the top export destination for country i , and turns off if the USA regains the top position. This switching definition reflects the empirical reality: four of the eight treated countries see the USA re-overtake China at some point during the sample period. Because the shock is plausibly exogenous to pre-trend voting behavior at UNGA, it is credible that it satisfies the identification assumptions. The figure with the estimated model includes a pre-trend fit, which resembles the treated behavior of vote similarity, providing evidence consistent with the key assumption of SDiD.

Predictors include per capita GDP from the Gravity Database (Gurevich and Herman 2018). Trade data are drawn from the International Trade and Production Database (Borchert et al. 2021). To

avoid double-counting, we use the value reported by the importing country to compute total trade between two dyads, as importers have a greater interest in accurately measuring this value for tax collection purposes (Borchert et al. 2021). We compute the share of total trade with China and the share of total trade with the United States as our two measures of trade. As our measure of power, we use the Global Power Index (GPI) from the Pardee Institute at the University of Denver. GPI combines demographic, economic, military, technological, and diplomatic dimensions. We calculate the power gap between all countries and the US as the difference between their GPI indices. We also include distance to the US, as countries closer to the US will have a bigger incentive to align their foreign policy with it. Another economic variable is the exchange rate, since many countries complain about China's devalued currency. A country with a more devalued currency is expected to be more aligned with China than one with a less devalued currency. We also include per capita income, as reported by the World Bank. Table 1 presents the descriptive statistics.

We further augment the SDiD covariate matrix with three institutional variables that capture political and economic alignment channels. First, we include a parliamentary system indicator, coded 1 for parliamentary regimes and 0 otherwise, from the Database of Political Institutions (Cruz, Keefer, and Scartascini 2021). Parliamentary systems may respond differently to trade shocks because the executive depends on legislative confidence, potentially moderating foreign-policy flexibility (Chen and Zhou 2021). Second, we include a military executive indicator, coded 1 when the chief executive has a military background, from the same source. Military executives may pursue distinct foreign-policy orientations, particularly regarding alignment with great powers (Morgan 2019). Third, we construct an indicator for the existence of a trade agreement with the United States (preferential trade agreement, free trade agreement, or economic integration agreement) from the Dynamic Gravity Database (Gurevich and Herman 2018). Countries with a US trade agreement face institutional constraints that may attenuate realignment toward China (Kastner and Pearson 2021; Flores-Macías and Kreps 2013). Because the DPI covers only up to 2015, we forward-fill institutional variables to later years, which is reasonable given the slow pace of regime-type changes.

A potential threat to our research design is the occurrence of a global economic crisis in 2008 (Frankel and Saravelos 2012). It is possible that Brazil may react differently in its foreign policy than other countries due to the financial crisis. To rule out the possibility that Brazil's alignment shift

Table 1: Descriptive statistics of panel variables.

Variable	N	Descriptive statistics
		N = 1,920 ^l
Treatment	1,920	8 (0.4%)
Absolute ideal point distance to China	1,920	0.91 (0.79) [0.00, 4.22]
Power Index	1,920	0.00 (0.50) [-0.12, 5.01]
Absolute ideal point distance to US	1,920	2.64 (0.92) [0.00, 5.06]
Share of trade with US	1,920	0.00 (0.50) [-0.41, 1.87]
Share of trade with China	1,920	0.00 (0.50) [-0.36, 5.08]
GDP (current US\$)	1,920	0.00 (0.50) [-0.37, 2.86]
Official exchange rate (LCU per US\$, period average)	1,920	0.00 (0.50) [-0.15, 6.82]
Physical distance to US	1,920	0.00 (0.50) [-1.16, 1.10]
Power Gap to US	1,920	0.00 (0.50) [-3.90, 0.43]
Current Account Balance as % of GDP	1,920	0.00 (0.50) [-3.32, 2.64]
Latin America Country	1,920	400 (21%)
Government Deficit as % of GDP	1,920	0.00 (0.50) [-6.68, 3.39]
Parliamentary System	1,920	748 (39%)
Military Executive	1,920	200 (10%)
Trade Agreement with US	1,920	159 (8.3%)

^ln (%); Mean (SD) [Min, Max]

Note: Unit = variable-specific (see row labels). Time window = 1990-2022 country-year panel used in the Brazilian SDiD setup. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

in 2009 was driven by macroeconomic turbulence rather than status salience, we augment our SDiD covariate matrix with two annual indicators capturing different facets of the 2008–09 crisis, both drawn from the Global Macro Database (Müller et al. 2025): a measure of current account balance (% GDP), to capture external sector stress and reliance on foreign financing; and a measure of Government Budget Deficit (% GDP), to reflect fiscal pressure and stimulus capacity.

With variables defined, we outline the estimation strategy next.

4 Descriptive Analysis

In the figure below, we plot the evolution of the UNGA ideal points for the US, Brazil, and China over time. As we can see, Brazil has been closer to China than to the US since the beginning of

the data series, in 1990, up to 2022, during Bolsonaro's government. Although Brazil moved away from China under Bolsonaro, Brazil was still closer to China than to the US. This is a testament to how much Brazil is structurally aligned with China at the UNGA.

It also shows that Brazil is moving more over time than China, which is mostly stable over time. Thus, this indicates that Brazil is moving closer to China than the opposite, as expected, given the power asymmetry between the two countries.

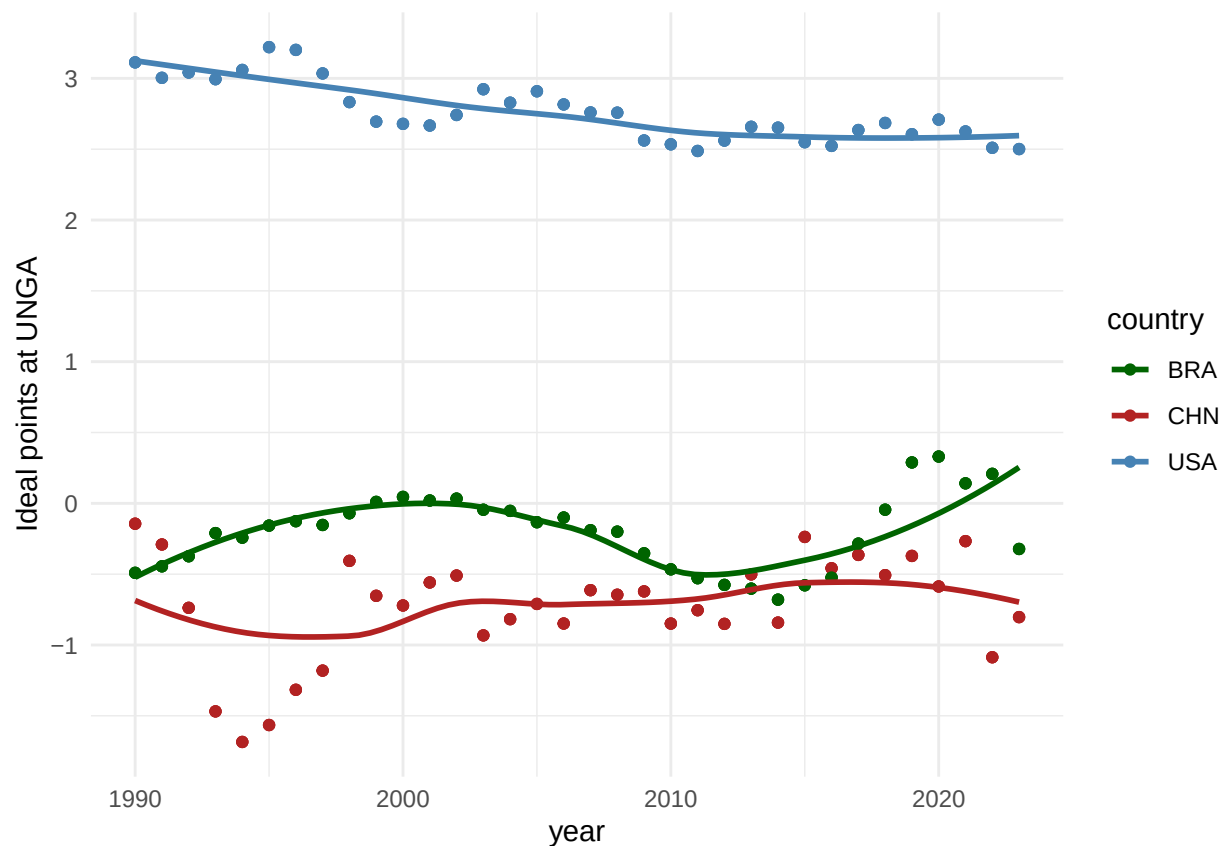


Figure 2: Evolution of UNGA ideal points for the United States, Brazil, and China over time. Note: Unit = UNGA ideal-point score (Bailey et al. scale). Time window = 1990-2022 annual series. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

Having established that it is mostly Brazil moving around rather than China, we can use as our main variable the absolute distance between Brazil's and China's ideal points, with the understanding that, apart from the beginning of the nineties, Brazil is almost always closer to the center than China and that Brazil is the one moving around.

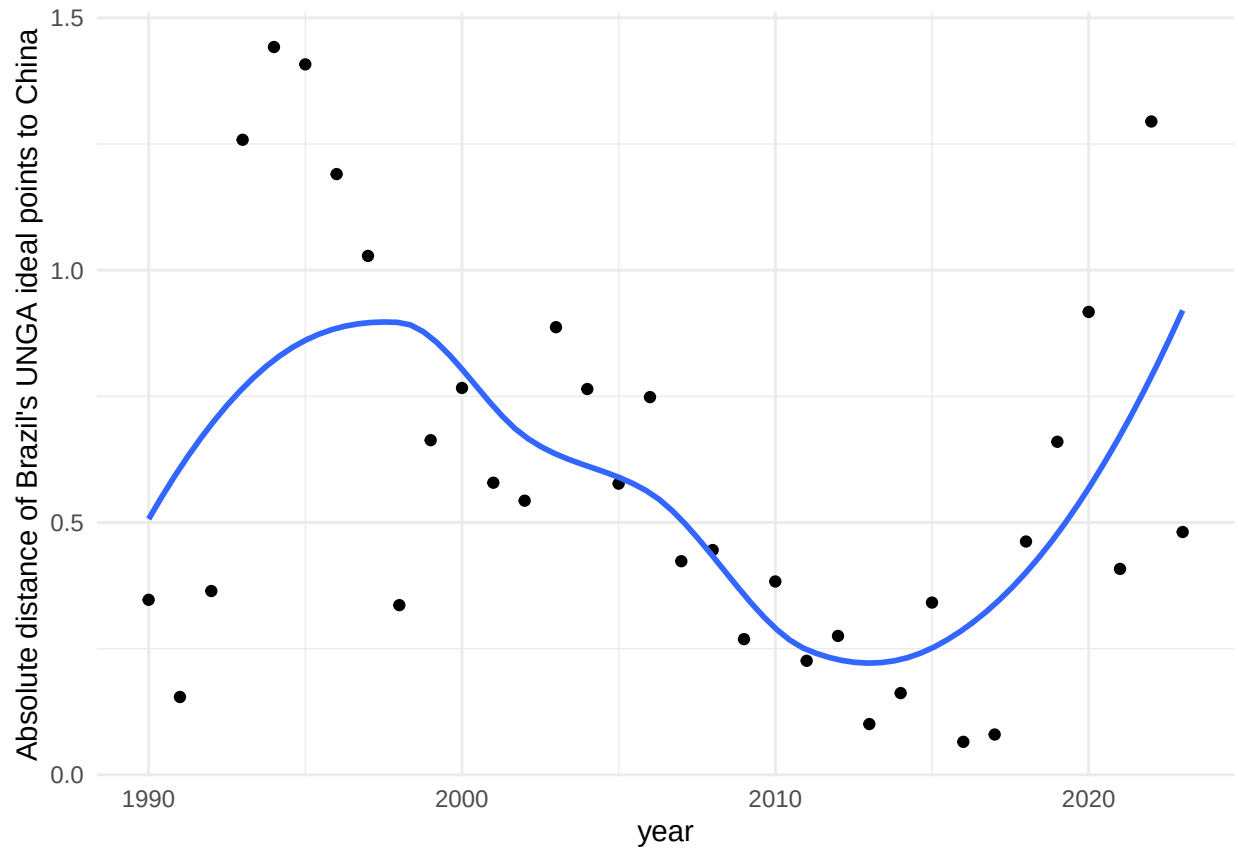


Figure 3: Absolute distance between Brazil and China in UNGA ideal points over time. Note: Unit = absolute UNGA ideal-point distance to China (points). Time window = 1990-2022 annual series. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

The question one wonders after seeing the graphs is: why was Brazil getting closer over time? The main hypothesis discussed in the literature is the gravitational pull of the Chinese economy, as indicated by the percentage of Brazil's exports that went to China during the considered period. The figure below shows that around the year 2000, the percentage of exports destined for China increased at a roughly constant rate, exhibiting a steep variation that does not exactly align with the evolution of political alignment presented earlier, although there is some positive correlation.

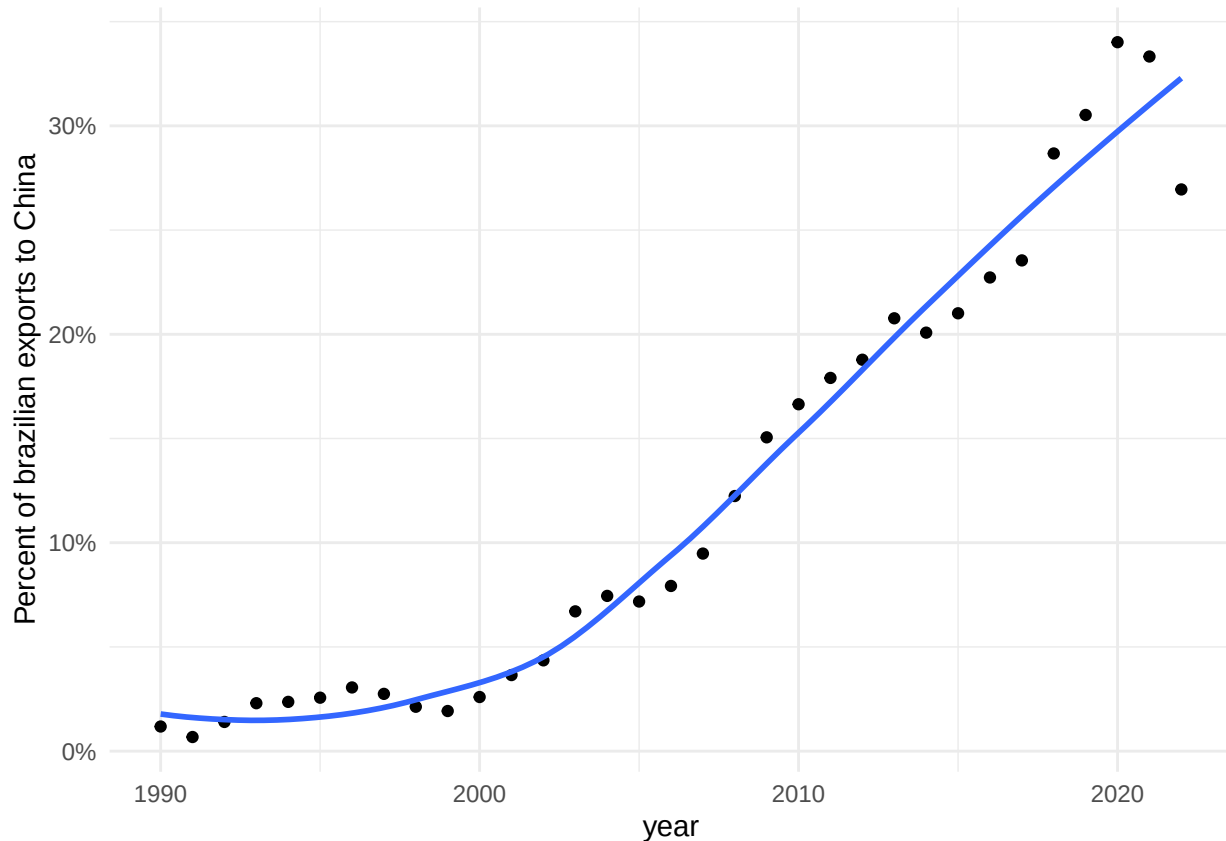


Figure 4: Share of Brazil's exports destined for China over time. Note: Unit = share of Brazil's exports to China (percent). Time window = 1990-2022 annual series. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

5 Empirical Results

The graph below presents the evolution of political alignment between Brazil and China at the UNGA and among a donor pool of countries before and after China became Brazil's main trade

partner. The counterfactual estimate from the donor pool allows us to estimate the causal effect of the treatment on the similarity of votes at UNGA. The estimate is -0.26 points, which means that becoming the main trade partner implied a decrease of 0.26 points in the absolute distance between the ideal points of Brazil and China after 2009. Relative to Brazil’s pre-treatment mean distance (0.647), this corresponds to a 40.8% reduction. The point estimate is -0.26 with a 95% CI of (-0.51, -0.02) and p-value 0.032 under placebo-based inference. To put the magnitude in perspective, the estimated reduction of 0.26 ideal-point units can be compared with other known drivers of UNGA alignment. He, Zheng, and Chen (2025) find that Chinese arms exports shift recipients’ ideal points toward China by approximately 0.08 points (2SLS), while Wheeler (2021) reports a comparable shift of 0.08 points from non-Western arms imports. Lektzian and Biglaiser (2023) estimate that the combination of US foreign aid and economic sanctions shifts bilateral voting similarity by 0.09–0.20 points. Our estimate of 0.26 points from a single trade rank reversal is thus comparable in magnitude to the effect of arms transfers and roughly half the effect of sustained US coercive diplomacy. This suggests that discrete status milestones can produce alignment shifts on par with material policy instruments that the literature has identified as consequential.

In Figure 5, we plot the results for Brazil, along with the top 10 controls, filtered by weight size. As we can see from the graph, no other country displays behavior similar to Brazil’s, which lends credibility to the estimation of the causal effect².

6 Robustness Checks

The target estimand of the SDiD estimator is the average treatment effect on the treated for the period post-treatment, whose weights are different from zero. This means that, in theory, it may be capturing the effect of other treatments that happened at a prior or later time. The main argument of the paper is that there is a distinctive quality in becoming the number one trade partner. If that is the case, then we should not find any effect when China became the second (in 2003, after overtaking Argentina and again in 2005, after briefly losing the second spot to Argentina in 2004), nor a few years after being the top one for a while.

Thus, we ran two pre-treatment in-time placebo tests, in which we pretended the true treatment

²A figure with the weights for top controls is in the appendix

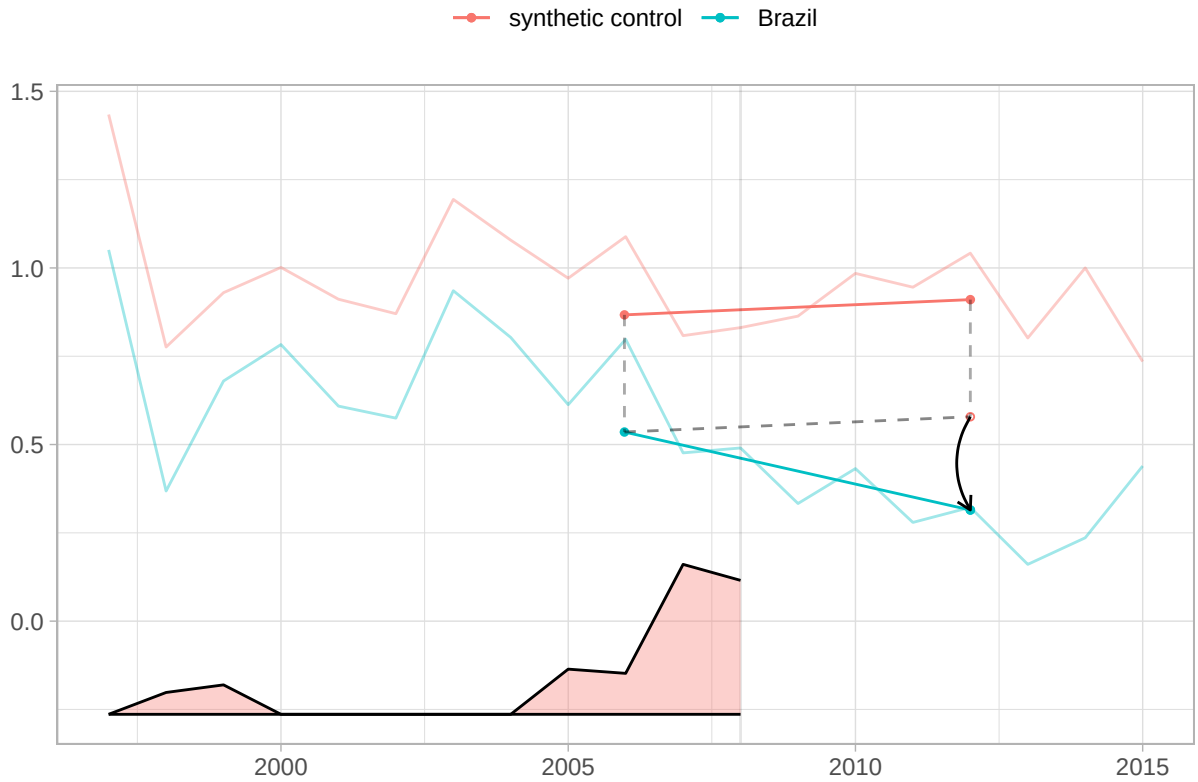


Figure 5: SDiD fit for Brazil and synthetic control with time weights. Note: Unit = absolute UNGA ideal-point distance to China (points). Standard error = placebo-based (reported in SDiD tables, not plotted here). Time window = pre-treatment and post-treatment years shown in the plot (treatment in 2009). Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

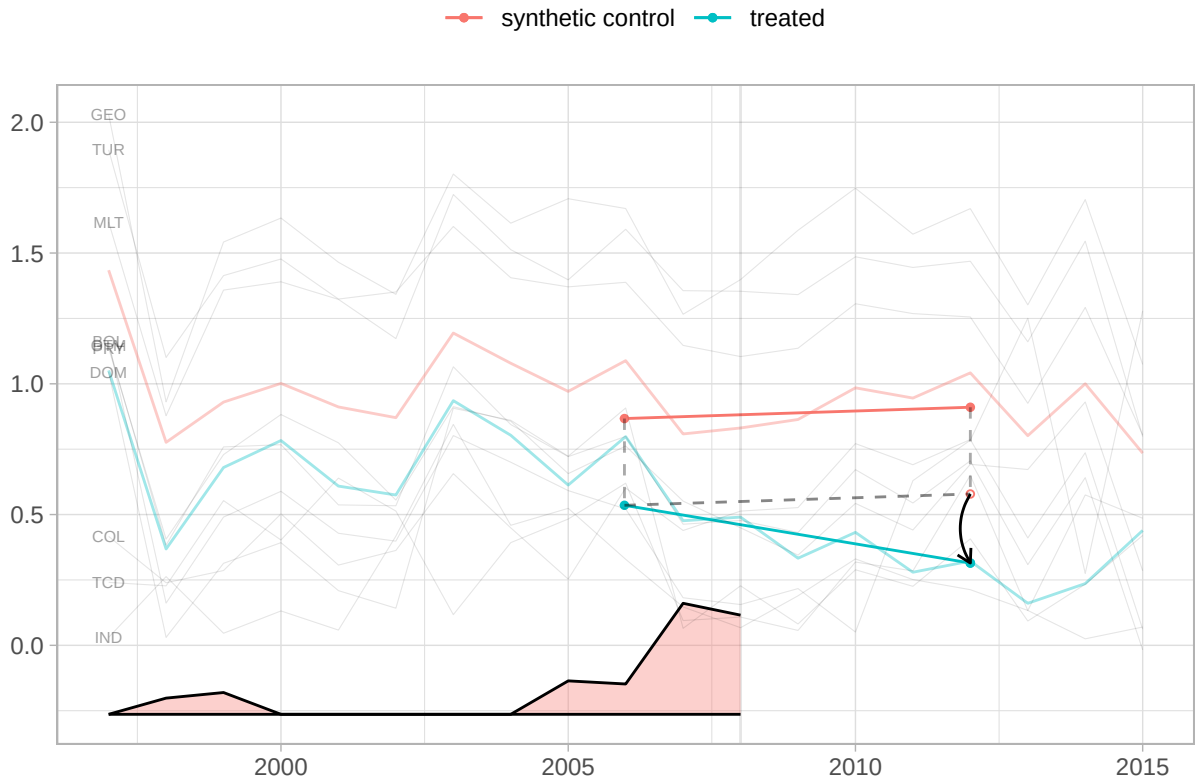


Figure 6: Top 10 donor controls by SDiD weight in the Brazilian specification. Note: Unit = country trajectories in absolute UNGA ideal-point distance; donor weights are unitless and sum to 1. Time window = pre-treatment and post-treatment years shown in the plot (treatment in 2009). Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

Table 2: Synthetic DiD: main model, pre-treatment placebos, and late-break falsification.

	(1) Main 2009	(2) Placebo 2003	(3) Placebo 2005	(4) Late-break 2012
ATT	-0.26 (0.12)	-0.12 (0.19)	-0.12 (0.13)	0.04 (0.14)
<i>Covariates:</i>				
Trade share China	✓	✓	✓	✓
Trade share US	✓	✓	✓	✓
Power index (GPI)	✓	✓	✓	✓
Power gap to US	✓	✓	✓	✓
Per-capita income	✓	✓	✓	✓
Exchange rate	✓	✓	✓	✓
Distance to US	✓	✓	✓	✓
HoG ideology	✓	✓	✓	✓
Current account (% GDP)	✓	✓	✓	✓
Gov. deficit (% GDP)	✓	✓	✓	✓
Parliamentary system	✓	✓	✓	✓
Military executive	✓	✓	✓	✓
US trade agreement	✓	✓	✓	✓
Sample truncated at	—	2008	2008	—

happened in 2003 and 2005. We also ran a post-treatment falsification at 2012 to probe an alternative explanation: that the estimated 2009 effect is actually driven by a later shock. For the two pre-treatment placebos, we limit the sample data to the year 2008, in order not to include the effect of the true treatment that started in 2009. The table below presents the results, listing the full set of covariates included in each specification. As we can see, the placebo effects are smaller and not statistically significant at the conventional 95% level.

Note: Unit = ATT in absolute UNGA ideal-point distance to China (points). Standard error = placebo-based (95 percent CI = estimate $\pm$ 1.96 $\times$ SE). Time window = main model uses treatment in 2009; pre-treatment placebos set pseudo-treatment at 2003 and 2005 with sample truncated at 2008; late-break falsification sets pseudo-treatment at 2012. Treatment definition = main model applies the treatment definition at 2009; placebo models redefine treatment at the pseudo-treatment year.

A potential confounder is the Lula da Silva presidency (2003–2010), which pursued an explicit “South-South” foreign policy reorientation. However, Lula took office in 2003—six years before the treatment. If his ideology were driving alignment with China, the pre-treatment placebos at

2003 and 2005 should show significant effects; they do not. Moreover, the cross-country analysis (Section 7) shows that the effect is not specific to Brazil or to any single leader, as it appears across countries with different political systems and leadership ideologies. The staggered treatment timing across countries rules out any single-country political confounder.

A potential concern is whether the estimated effect reflects the discrete change in status by the rank reversal or simply a non-linear function of trade growth. Three pieces of evidence address this concern. First, our SDiD specification includes the share of trade with China and the share of trade with the United States as time-varying covariates, absorbing the linear effect of trade accumulation on alignment. Second, the placebo tests at 2003 and 2005 are directly informative: in both years, Brazil’s trade with China was growing rapidly – China overtook Argentina to become Brazil’s second-largest partner in 2003 – yet neither placebo produces a significant effect. If alignment responded smoothly to trade growth, these years of steep increases should show comparable shifts; they do not. Third, the cross-country analysis exploits staggered treatment timing: different countries reached the rank reversal at different trade-share levels and in different years, providing variation that separates the rank effect from any single trade trajectory³. Together, these results support the interpretation that the status change itself – not the underlying trade growth – drives the observed alignment shift.

6.1 Salience in the media

If the salience channel is relevant, there should be concrete evidence in Brazilian media coverage. To test this, we collected all headlines, stories, op-eds, and editorials mentioning “China” in *Folha de São Paulo*, Brazil’s widest-circulation newspaper (*Folha de S.Paulo* 2016), from 2000 to 2014. Among 48,029 pieces, we identified 14,589 with “China” (or variants like “Chinese”) in the headline, quantifying the evolution of China’s salience in national discourse.

The stories encompass a wide range of themes, including sports, natural disasters, politics, economics, science, and more. However, over time, news pieces about China have increasingly reflected its growing relevance as Brazil’s largest trade partner since 2009.

³Adding higher-order polynomial terms of trade shares (quadratic, cubic) is not advisable in this context, as Gelman and Imbens (2019) show that polynomial specifications produce noisy estimates with poor confidence-interval coverage, a concern that applies a fortiori in an underpowered single-treated-unit design.

The evidence shows that, after 2009, Brazilian media attention to China not only increased but also shifted from general-interest stories to trade headlines. Figure 7 shows that news pieces about China peaked during the 2008 Beijing Olympics and remained high as China surpassed the US as Brazil’s largest trade partner. The shock year (2009) saw nearly twice the annual average of the pre-2008 period.

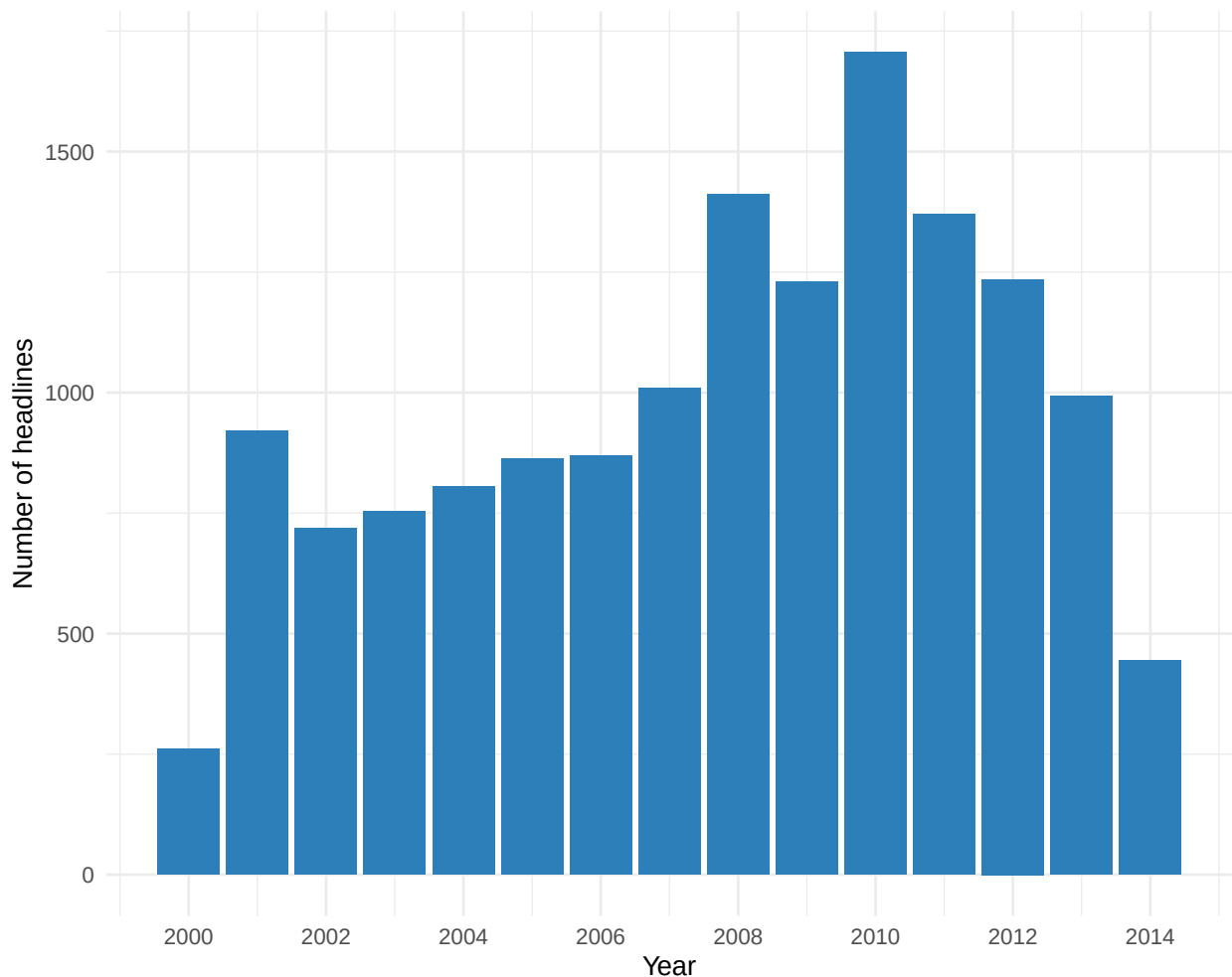


Figure 7: Number of Folha de Sao Paulo headlines mentioning China by year. Note: Unit = count of headlines. Standard error = not applicable. Cluster = not applicable. Time window = 2000-2014 annual series. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil’s top export destination displacing the USA, and 0 before 2009.

However, it is possible that China’s rise in the news is due to events unrelated to trade, such as increased tensions with the US or the Beijing Olympics. To clarify whether trade-related news became more salient after 2009, we utilized the OpenAI API for ChatGPT, model gpt-4.1-mini, created on April 10, 2025, to categorize the news—see the appendix for details—into 9 categories.

This step allows us to assess precisely how trade-focused news evolved over time.

This distinction is crucial because volume alone could reflect any China-related event. What matters is the substance of what journalists discussed. To track this, Figure 8 plots the share of trade-economy pieces in all China headlines. The peak is in 2004, when there was less news about China in total and it was the year that Brazil recognized China as a market economy at the WTO (14% of all trade-related stories that year), with mostly negative headlines. Manual reading of the trade news pieces at the time shows that a plurality of them are about soy import barriers put in place by China (38% of all trade-related stories that year). So, although it was a highly salient year, it was not the type of salience that would trigger pressure on foreign policy from the business community toward China.

A closer look at the evolving themes further illustrates this shift. Up to 2006, the main China-related coverage focused on health issues, North Korea and nuclear topics, as well as certain sports themes (“seleção brasileira”). By 2009, those subjects remained, but news about Asia, the China and Hong Kong stock markets, and, centrally, Brazilian company Vale do Rio Doce’s exports to China—closely linked with the Dollar and Brazilian trade—became prominent. This evidence is central to our proposed mechanism, as it illustrates the qualitative transformation in the economic relationship between Brazilian companies, the broader economy, and China. It is helpful to zoom in and compare 2008 with 2009 for further clarity.

A lexical snapshot emphasizes the qualitative break between years. By examining the most frequent trigrams, we gain insight into the changing content from year to year. Trigrams are strings of three tokens that help segment sentences into key phrases (Violos et al. 2018). In 2008, the most common three-word bundles included ‘earthquake’, ‘Olympic torch’, and ‘bird flu’. In 2009, they shifted decisively to ‘record iron-ore exports’, ‘Bovespa rises’, and ‘Brazil-China trade hits record’—precisely the salience cues our theory predicts.

Headlines with “bate” (“reach”) illustrate the pattern: In 2009, headlines like “Agribusiness exports reach record high, with China as the main highlight” and “China hits record iron ore imports” repeatedly appeared, while in 2008, “bate” appeared only once in an agribusiness context⁴.

⁴We used ChatGPT 4o to assist in the headlines translation from Portuguese into English. See <https://chatgpt.com/share/6862b919-4630-800e-ab9c-df9dee5ae56c>

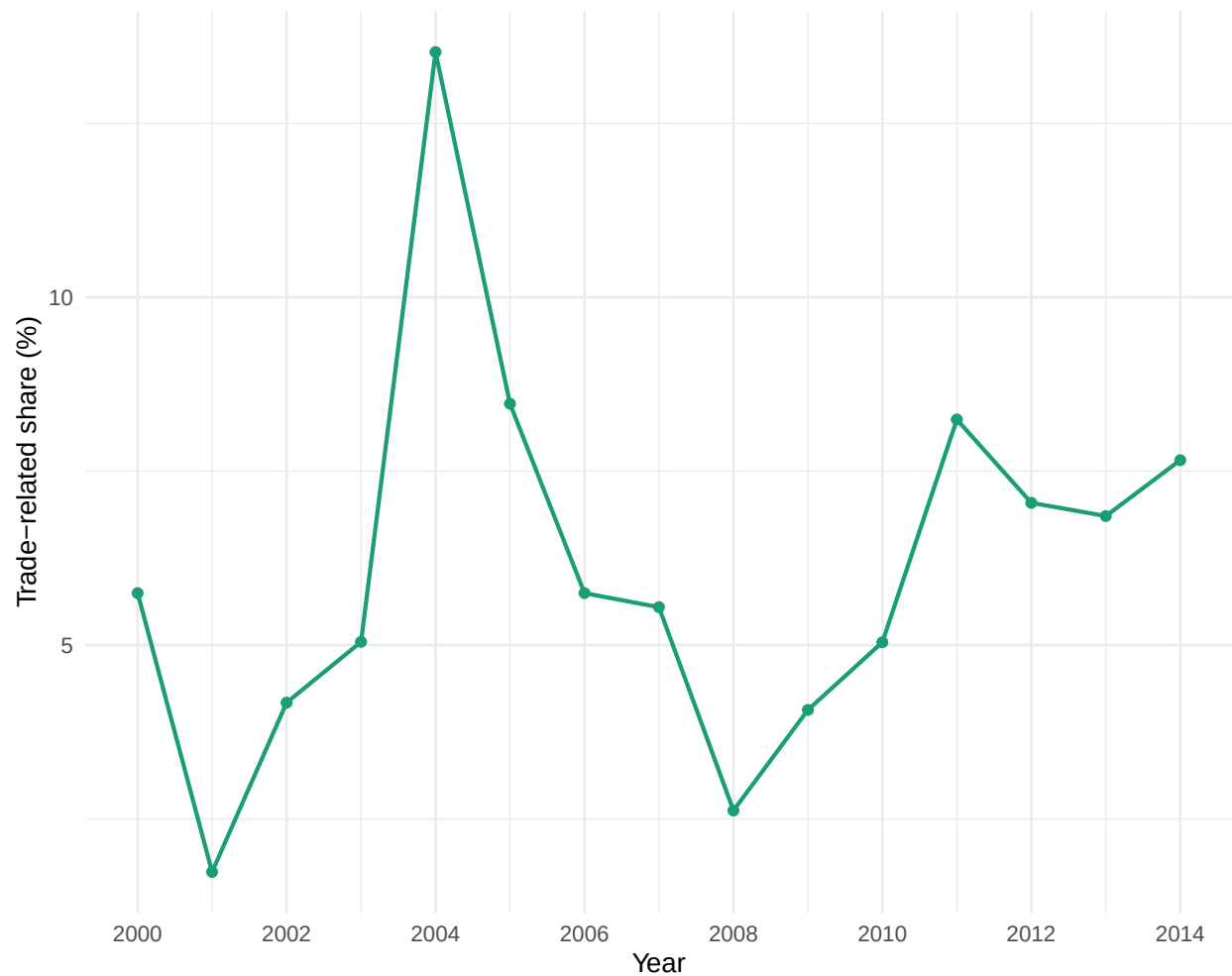


Figure 8: Share of trade-related items among all China headlines per year. Note: Unit = share of trade-related China headlines (percent). Standard error = not applicable. Cluster = not applicable. Time window = 2000-2014 annual series. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

Political elite discourse mirrored the media. During his May 2009 visit to Beijing, President Lula opened a business seminar by noting that ‘China has become, in 2009, Brazil’s largest trading partner’ and immediately linked this milestone to a coincidence of positions in international forums (Presidência da República Federativa do Brasil 2009). Two years later, Deputy Aldo Rebelo reminded the Foreign Affairs Committee that China’s elevation to first place justified deeper legal cooperation (Câmara dos Deputados 2011).

Taken together, media salience shifted in both volume and content precisely when China overtook the United States. Figure 7 documents a sustained increase in China-related headlines after 2009. Figure 8 shows that the share of trade-economy stories doubled immediately following the rank change – a pattern absent in prior years – while the lexical comparison (Figure 9) confirms the pivot from disaster and sports trigrams in 2008 to ‘record exports’, ‘iron ore’, and ‘Bovespa’ in 2009. The timing aligns with the SDiD breakpoint (2009), and pre-treatment placebo years with similar trade growth but no rank change show no comparable media pivot. These patterns are consistent with a media agenda shock centered on the economic relationship, as predicted by coarse-categorization theories: once China became Brazil’s largest partner, editors and readers re-sorted China stories into the ‘top-trade-partner’ category, amplifying their news value. This pattern is more consistent with a salience mechanism than with gradual interdependence. Media evidence is consistent with mechanism implications.

7 Cross-Country Evidence

The Brazilian case generated the theory and provided internally valid causal evidence via SDiD. However, it remains a single case, and we cannot rule out idiosyncratic confounders. We now test the generalizability of the status-salience theory using a cross-country design, which constitutes an independent out-of-sample test.

A natural question is whether the same effect occurs when China displaces the United States as the top trade partner in other countries. Our theory predicts effects specifically when an emerging power displaces the hegemon – not when China overtakes Belgium or Singapore. This distinction is theoretically motivated: the status shock is most salient when the displaced partner is the incumbent great power, whose position in a country’s trade hierarchy carries symbolic and geopolitical weight.

An important empirical feature of the cross-country sample is that the treatment is not absorbing: four of the eight treated countries experience treatment reversals – the USA regains the top position after China held it temporarily. This switching pattern makes standard absorbing-treatment estimators inappropriate and motivates our use of the *fect* IFE estimator (Liu, Wang, and Xu 2024), which natively accommodates treatment switching.

7.1 Main result (*fect* IFE)

We construct a panel of 99 countries where the United States was the top export destination at some point in the sample period. Among these, 59 countries experienced China displacing the US as the top partner at least once (treated observations), while the remaining 40 never experienced such a displacement (control group). The treatment indicator D_{it} equals 1 in any year China is currently the top export destination and switches off when the USA regains that position. This design accommodates the switching pattern and uses all available variation.

We estimate the ATT using the counterfactual estimator with interactive fixed effects (IFE) of Liu, Wang, and Xu (2024). The IFE model relaxes the strict parallel trends assumption by allowing for latent common factors with heterogeneous loadings, and cross-validation selects $r^* = 1$ factors. The overall ATT is -0.074 (bootstrap SE = 0.029, 95% CI [-0.132, -0.016], $p = 0.012$), indicating that when China holds the top trade partner position, countries move significantly closer to China in UNGA voting. Substantively, this corresponds to about 9.4% of the pre-treatment mean distance to China. For comparison, the cross-sectional standard deviation of UNGA distance to China across all countries in the panel is approximately 0.79 points, so the IFE estimate represents roughly one-tenth of a standard deviation – a modest but meaningful shift when considering that it is produced by a single symbolic milestone rather than a structural change in economic ties.

The table below compares results across estimators. The *fect* FE model (two-way fixed effects without latent factors) yields a similar estimate. The Callaway and Sant’Anna (2021) absorbing-treatment estimator, which can only use the 7 countries where China never lost the top position, also produces a directionally consistent result, confirming robustness.

Note: Unit = ATT in absolute UNGA ideal-point distance to China (points). Standard error = bootstrap (*fect*) / analytical (C&S). Cluster = country level. Time window = panel covers 1990-

Table 3: Cross-country ATT estimates: fect (switching treatment) vs. C&S (absorbing-only).

Estimator	ATT	SE	p-value
fct FE (switching)	-0.026	0.028	0.343
fct IFE, $r^*=1$ (switching)	-0.074	0.029	0.012
C&S absorbing-only (7 countries)	-0.117	0.045	0.009

2023. Treatment definition = Treatment indicator equals 1 in any year China is currently the top export destination; turns off if USA regains the position (switching treatment).

Figure 9 presents the consolidated diagnostic tests following Liu, Wang, and Xu (2024). Panel (a) shows the IFE dynamic treatment effects: pre-treatment estimates fluctuate around zero, and the effect turns negative upon treatment entry, peaking in the first few years. Panel (b) displays the FE placebo test, which assesses whether the model correctly predicts pre-treatment outcomes. The conventional test does not reject the null of zero placebo effects ($p = 0.139$), though with only 59 treated countries, statistical power is limited. More informatively, the equivalence test – which reverses the burden of proof by testing the null that pre-treatment effects *are* meaningfully large – rejects at the 5% level ($p = 0.04$), providing positive evidence that pre-treatment effects are negligibly small relative to the ATT (Chiu et al. 2025). Panel (c) shows the carryover test: the conventional test does not reject the null of zero carryover effects ($p = 0.518$), and the equivalence test rejects the null that carryover effects are meaningfully large ($p = 0.014$), confirming that the alignment effect dissipates after exit. The equivalence test plots are presented in the Appendix.

7.2 Exit effects and treatment reversals

A distinctive advantage of the switching-treatment framework is the ability to estimate exit effects – what happens to UNGA alignment when the USA regains the top trade partner position and the treatment turns off. This provides a direct test of the salience mechanism: if the alignment shift is driven by the ongoing salience of China’s status as top partner, the effect should dissipate when that status is lost.

The exit-aligned gap plot shows that the treatment effect dissipates after the USA regains the top position – consistent with the carryover test results reported in Figure 9(c). This pattern provides direct evidence for the salience mechanism (H4): unlike structural explanations that predict persis-

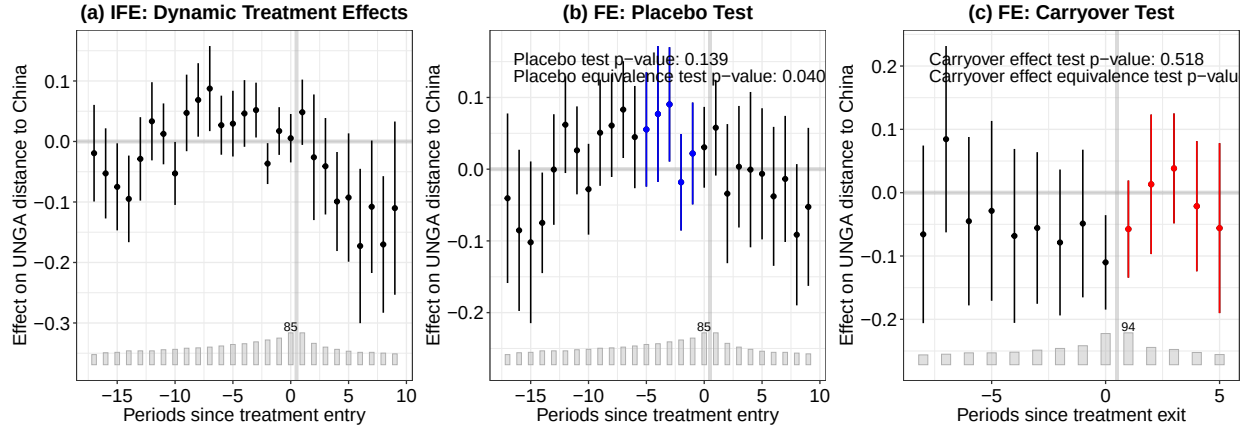


Figure 9: Diagnostic tests for the cross-country fect analysis (Liu, Wang & Xu 2024). (a) IFE dynamic treatment effects; (b) FE placebo test with placebo periods in blue; (c) FE carryover test with carryover periods in red. Note: Unit = ATT in absolute UNGA ideal-point distance to China (points). Standard error = bootstrap (500 replications). Time window = 1990-2023 panel. Treatment definition = Treatment indicator equals 1 in any year China is currently the top export destination; turns off if USA regains the position (switching treatment).

tent effects regardless of current trade status, the salience channel predicts that the effect is tied to the cognitive category “China = top partner” and should fade when that category is disrupted.

We complement the fect analysis with PanelMatch (Imai, Kim, and Wang 2023), which separately estimates the Average Treatment Effect on the Treated (ATT, switch-on) and the Average Treatment Reversal Effect on the Treated (ART, switch-off).

The PanelMatch results are noisier than fect, as expected with matching-based inference on a small panel. The ATT estimates are small at lead 0 (0.012) but become progressively negative at longer leads, consistent with a gradual alignment effect. The ART at lead 0 is 0.059 (95% CI [-0.046, 0.165]), directionally consistent with countries moving away from China when the USA regains the top position. While the individual lead-level estimates are imprecise due to the small number of switching events, the overall pattern – negative entry effects that build over time and positive initial exit effects – is consistent with the fect results and supports a mechanism tied to current trade status.

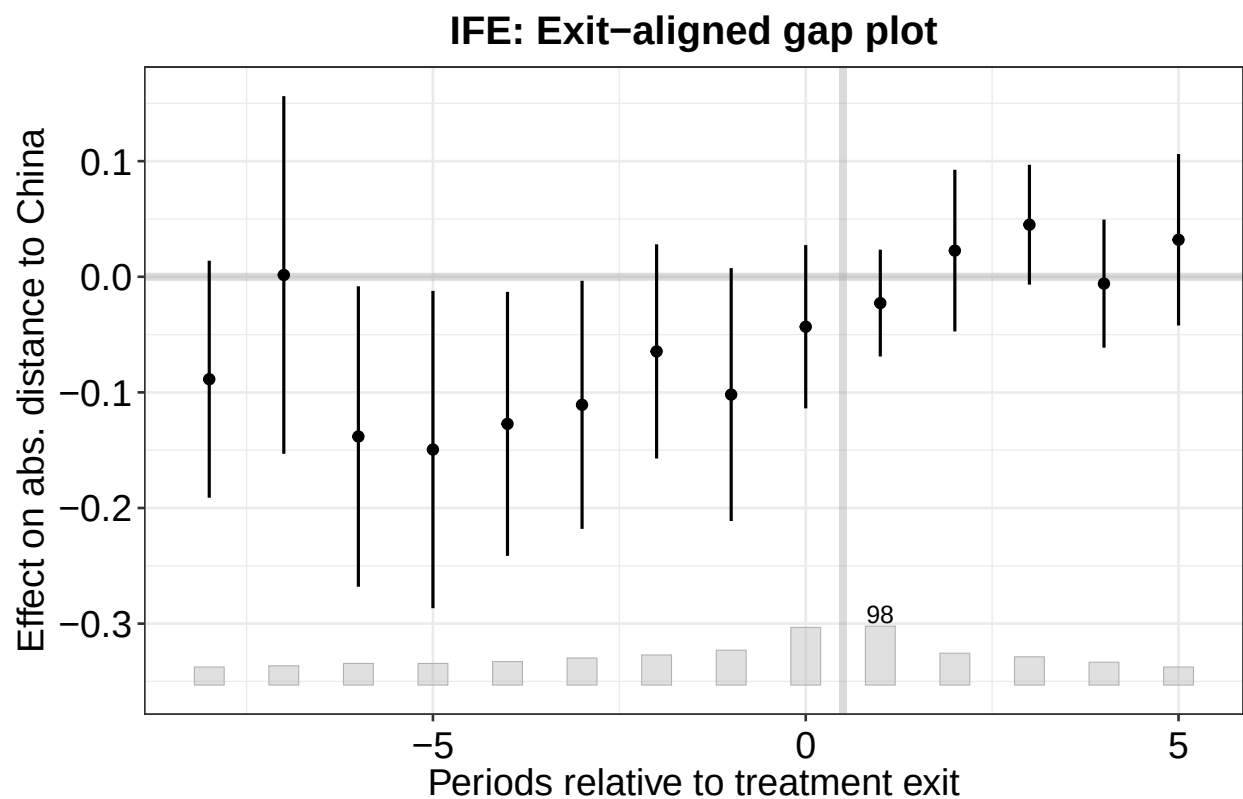


Figure 10: IFE: Exit-aligned gap plot showing what happens when the USA regains the top trade partner position. Note: Unit = ATT in absolute UNGA ideal-point distance to China (points). Standard error = bootstrap (500 replications). Time window = 1990-2023 panel. Treatment definition = Treatment indicator equals 1 in any year China is currently the top export destination; turns off if USA regains the position (switching treatment).

PanelMatch: Entry vs. Exit Effects

Imai, Kim & Wang (2023) -- lag = 1, no refinement

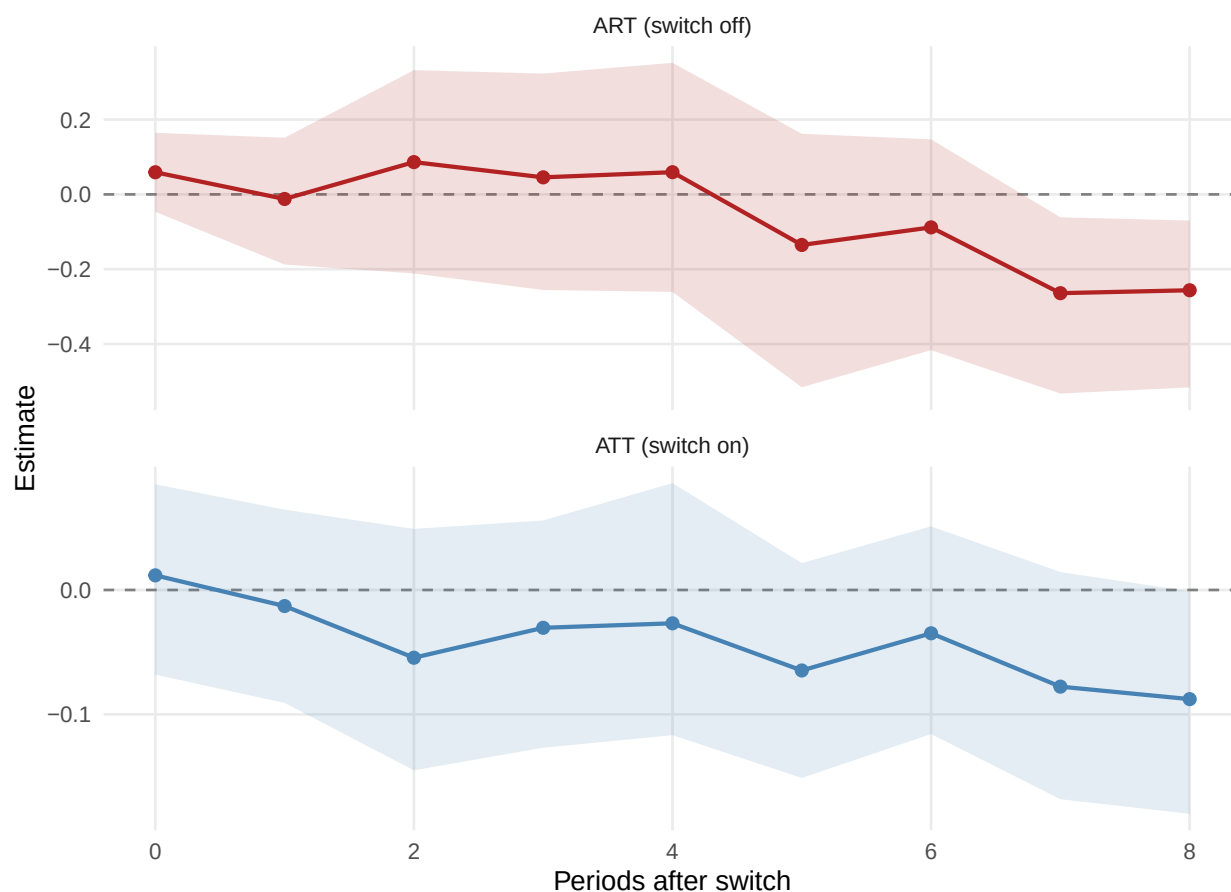


Figure 11: PanelMatch: Entry (ATT) and exit (ART) effects of China's status as top trade partner. Note: Unit = ATT/ART in absolute UNGA ideal-point distance to China (points). Standard error = bootstrap (1000 iterations). Time window = 1990-2023 panel. Treatment definition = Treatment indicator equals 1 in any year China is currently the top export destination; turns off if USA regains the position (switching treatment).

7.3 Raw data

To provide transparency on the treated countries driving the cross-country result, we plot the raw UNGA voting distance to China for each treated country. Each panel shows the annual absolute distance to China in UNGA ideal points, with a LOESS smoother and a vertical dashed line marking treatment periods.

The raw data reveal that most treated countries experienced a decline in ideological distance to China around the time of the rank reversal, consistent with the estimated ATT. The raw trajectories also confirm the switching pattern: countries where the USA regains the top position show a partial rebound, consistent with the exit effects estimated above.

7.4 Dynamic treatment effects and the temporality of the status effect

The effect gap plot above already displays the dynamic treatment effects relative to treatment entry. The effect emerges sharply at the time of rank reversal, peaks in the first few years post-treatment, and then gradually attenuates. This temporal pattern is substantively important.

A status-salience mechanism, by definition, operates through a cue – the rank reversal – that is most prominent when it first occurs. Over time, the novelty fades: the new status quo becomes normalized, media attention moves to other stories, and policymakers' consideration sets shift. The gradual attenuation is therefore theoretically predicted: unlike mechanisms that depend on durable structural changes (such as trade volume or economic dependence), the salience channel should produce a temporary effect that dissipates as the cue loses its informational novelty. This is what the estimates suggest.

The exit-aligned analysis provides a more direct test of temporality than the entry-aligned event study alone. The Brazilian case already hinted at the attenuation pattern, as the raw UNGA distance data showed a rebound after 2015, but the single-case design could not distinguish the rebound from idiosyncratic noise (e.g., the transition from Rousseff to Temer after an impeachment). The cross-country exit effects confirm that the rebound is systematic across treated countries, not a Brazil-specific artifact⁵.

⁵We thank an anonymous reviewer for raising the possibility that the effect might dissipate over time.

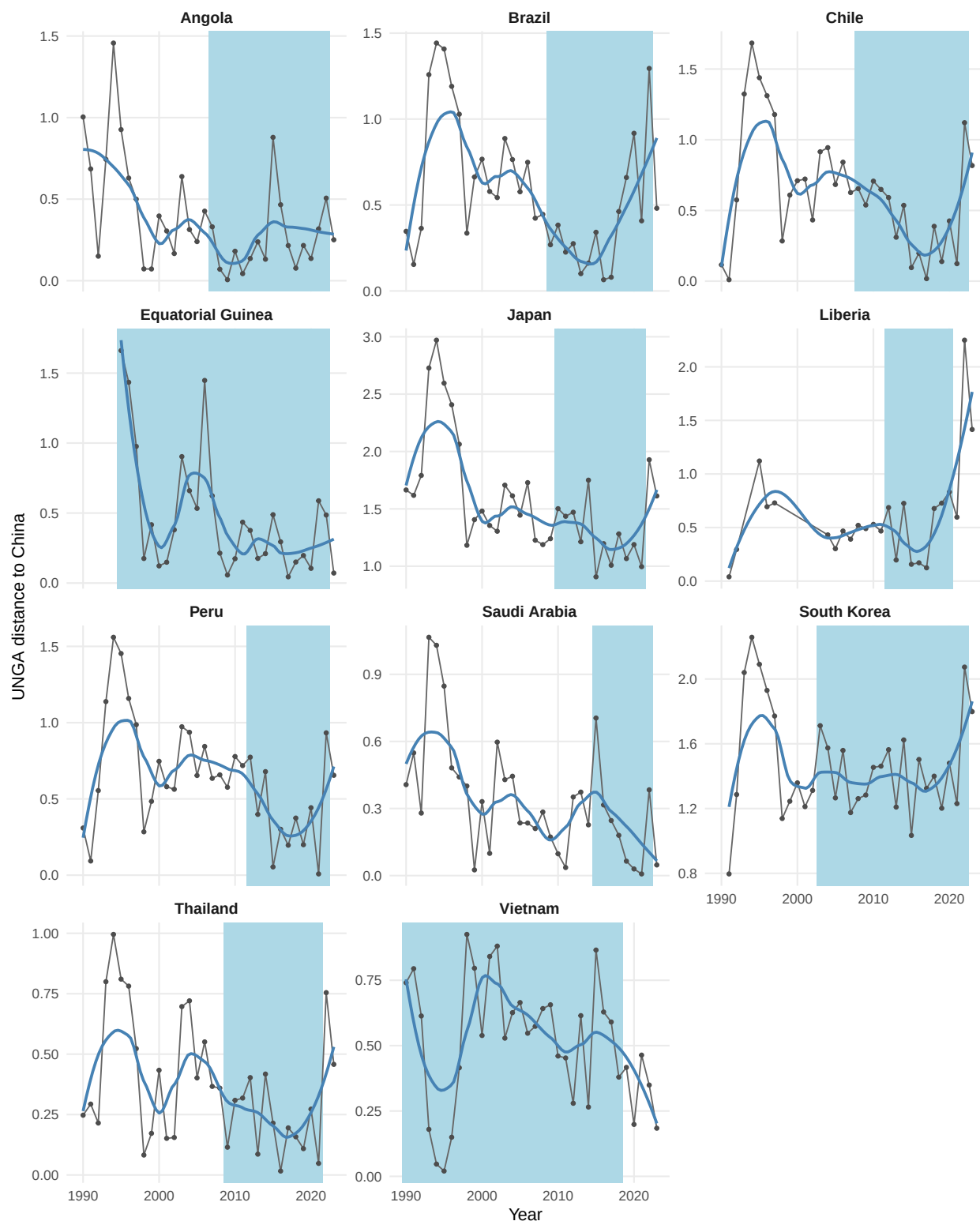


Figure 12: UNGA voting distance to China for treated countries where China displaced the USA. Note: Unit = absolute UNGA ideal-point distance to China (points). Time window = treated-country yearly series within 1990-2022; treatment year varies by country. Treatment definition = Treatment indicator equals 1 in any year China is currently the top export destination; turns off if USA regains the position (switching treatment).

7.5 Alternative Explanations

We systematically consider the most plausible alternative explanations and their empirically distinguishable predictions:

Alternative	Key prediction	Discriminating evidence
Structural bandwagoning	Persistent alignment shift	Entry-aligned attenuation + exit effects show dissipation (H4)
Chinese coercion	Effect for any displaced partner	Only hegemon displacement produces effect (H2)
Gradual trade growth	Monotonically increasing effect	Discrete effect; Carmody, Dasandi, and Mikhaylov (2020) show continuous trade growth produces hedging, not alignment
Ideologically aligned leader	Step-change at leader inauguration	DPI controls; Dreher and Jensen (2013) find leader change pushes votes toward the US, which is the opposite direction
2008 financial crisis	Temporal confound in 2008–2009	Staggered treatment: rank reversals in other years also produce effect; effect absorbs common shocks via IFE

The exit effects provide our most powerful discriminating evidence. Nearly all alternative mechanisms predict persistent effects: if structural bandwagoning, coercion, or accumulated economic interests drive alignment, the effect should not reverse when the USA regains the top position. The effect exit plot and PanelMatch ART show precisely this reversal – a pattern uniquely consistent with a salience mechanism, where the effect is tied to the cognitive category “China = top partner” and

dissipates when that category is disrupted.

We do not claim to have ruled out all alternatives. The effect of ideologically aligned leaders is partially testable but not completely eliminable absent exogenous leader variation. Reciprocal diplomatic concessions are largely unobservable. What we show is that the empirical signature – a discrete, hegemon-specific effect that dissipates upon exit – is most consistent with the status-salience mechanism and least consistent with the leading alternatives.

8 Conclusion

We have proposed a theory that status change induced by trade rank reversal, when China surpassed the US as the main trade partner of a country, can produce foreign policy realignment. We found initial support for this effect when China overtook the US as Brazil’s leading trade partner. The evidence is consistent with the idea that the symbolic significance of overtaking a historic trade partner prompted a substantive shift in diplomatic posture, beyond changes in trade volume. The cross-country regressions confirm the original finding, although with a smaller magnitude, which is expected, since Brazil is the most likely case. At the same time, evidence on the underlying salience mechanism is correlational and suggestive rather than causally identified.

Overall, the results suggest that we should look beyond the increasing trade ties between countries, as has been done in the literature so far (Yan and Zhou 2021; Flores-Macías and Kreps 2013; Reilly 2013), and also focus on qualitative shifts in the public perception of a trade partner’s status.

The results have broad implications for the literature on status. While it has focused on explaining how the search for status may trigger conflicts, it has overlooked robust empirical evidence that an increase in status changes the behavior of another country. Thus, contrary to Mercer (2017), who argued that prestige is illusory, we have shown that it matters in a realm important for a country like China, namely, voting in the UNGA.

Our study has some limitations that future research could investigate further. The cross-country analysis confirms that the status effect generalizes among countries where China displaced the United States, but we cannot yet test whether similar effects arise when China overtakes other partners (e.g., former colonial powers or regional trade leaders), as the number of such cases re-

mains small and heterogeneous. Furthermore, we lack cross-country media data to verify whether the salience mechanism operates in the same way outside Brazil.

Secondly, the proposed mechanism has only indirect evidence. We have shown that newspaper coverage of trade ties with China shifted when it surpassed the US as the primary trade partner. However, we lack evidence on how business lobbies responded to it, and we do not have systematic data on the public or political elites. Qualitative studies with political and economic elites could attempt to recover their thinking at the time, for instance. Laboratory experiments could be conducted to provide evidence on the psychological and cognitive mechanisms underlying status based on a change in the rank of a partner.

Our findings also carry practical implications. From the perspective of US policymakers, the results suggest that symbolic milestones in trade hierarchies deserve attention alongside aggregate trade volumes. If losing the top trade partner position triggers a discrete shift in diplomatic alignment, then monitoring and responding to these threshold crossings – rather than focusing solely on trade balances – may be a more effective strategy for maintaining influence. More broadly, the finding that alignment effects dissipate when the USA regains the top position implies that trade-status losses are not irreversible, and that targeted trade engagement could mitigate diplomatic realignment.

Despite these limitations, our study is the first to provide evidence for such trade-status effects in international relations. By highlighting trade-status salience, this study also opens up new avenues for understanding how discrete milestones in economic ties can reshape international alignments and invites extensions to other domains (e.g., FDI, foreign aid, loans) and regions.

References

- Aiyar, Shekhar, Davide Malacrino, and Andrea F. Presbitero. 2024. “Investing in Friends: The Role of Geopolitical Alignment in FDI Flows.” *European Journal of Political Economy* 83 (June): 102508. <https://doi.org/10.1016/j.ejpoleco.2024.102508>.
- Anagol, Santosh, and Thomas Fujiwara. 2016. “The Runner-up Effect.” *Journal of Political Economy* 124 (4): 927–91.
- Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager.

2021. “Synthetic Difference-in-Differences.” *American Economic Review* 111 (12): 4088–118. <https://doi.org/10.1257/aer.20190159>.
- Baekgaard, Martin, Julian Christensen, Casper Mondrup Dahlmann, Asbjørn Mathiasen, and Niels Bjørn Grund Petersen. 2019. “The Role of Evidence in Politics: Motivated Reasoning and Persuasion Among Politicians.” *British Journal of Political Science* 49 (3): 1117–40. <https://doi.org/10.1017/S0007123417000084>.
- Bai, Jushan. 2009. “Panel Data Models with Interactive Fixed Effects.” *Econometrica* 77 (4): 1229–79. <https://doi.org/10.3982/ECTA6135>.
- Bailey, Michael A, Anton Strezhnev, and Erik Voeten. 2017. “Estimating Dynamic State Preferences from United Nations Voting Data.” *Journal of Conflict Resolution* 61 (2): 430–56.
- Bastardo, Nicolas, Michael J. Matthews, Gwendolin B. Sajons, Tyler Ransom, Thomas K. Kelemen, and Samuel H. Matthews. 2023. “Instrumental Variables Estimation: Assumptions, Pitfalls, and Guidelines.” *The Leadership Quarterly* 34 (1): 101673. <https://doi.org/10.1016/j.leaqua.2022.101673>.
- BBC News. 2021. “China Overtakes US as EU’s Biggest Trading Partner.” <https://www.bbc.com/news/business-56093378>.
- Blackwell, Matthew, and Adam N Glynn. 2018. “How to Make Causal Inferences with Time-Series Cross-Sectional Data Under Selection on Observables.” *American Political Science Review* 112 (4): 1067–82.
- Borchert, Ingo, Mario Larch, Serge Shikher, and Yoto V Yotov. 2021. “The International Trade and Production Database for Estimation (ITPD-E).” *International Economics* 166: 140–66.
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer. 2022. “Salience.” *Annual Review of Economics* 14: 521–44. <https://doi.org/10.1146/annurev-economics-051520-011616>.
- Callaway, Brantly, and Pedro H. C. Sant’Anna. 2021. “Difference-in-Differences with Multiple Time Periods.” *Journal of Econometrics* 225 (2): 200–230. <https://doi.org/10.1016/j.jeconom.2020.12.001>.
- Câmara dos Deputados. 2011. “Comissão Aprova Acordo de Direitos Civil e Comercial Entre Brasil e China.” <https://www.camara.leg.br/noticias/222695-comissao-aprova-acordo-de-direitos-civil-e-comercial-entre-brasil-e-china/>.
- Carmody, Pádraig, Niheer Dasandi, and Slava Jankin Mikhaylov. 2020. “Power Plays and Balanc-

- ing Acts: A New Mapping of the African Geoeconomic Landscape in Response to the Rise of China.” *Political Studies* 68 (4): 852–70. <https://doi.org/10.1177/0032321719882791>.
- Chen, Qin, and Yi Zhou. 2021. “Whose Trade Follows the Flag? Institutional Constraints and Economic Responses to Bilateral Relations.” *Journal of Peace Research* 58 (6): 1207–23. <https://doi.org/10.1177/0022343321992825>.
- Chiu, Albert, Xingchen Lan, Ziyi Liu, and Yiqing Xu. 2025. “Causal Panel Analysis Under Parallel Trends: Lessons from a Large Reanalysis Study.” *American Political Science Review*. <https://doi.org/10.1017/S0003055424001035>.
- Conlon, John J. 2025. “Attention, Neglect, and Persuasion.” https://johnjconlon17.github.io/website/conlon_attention_persuasion.pdf.
- Coutinho, Yuri Bravo, and Júlio César Cossio Rodriguez. 2024. “Chinese Double Effect on Brazilian Foreign Policy (2003-2018).” *Contexto Internacional* 46 (2): e20230014. <https://doi.org/10.1590/s0102-8529.20244602e20230014>.
- Cruz, Cesi, Philip Keefer, and Carlos Scartascini. 2021. “Database of Political Institutions 2020.” *Inter-American Development Bank Research Department*. <https://doi.org/10.18235/0003049>.
- Dafoe, Allan, Jonathan Renshon, and Paul Huth. 2014. “Reputation and Status as Motives for War.” *Annual Review of Political Science* 17 (1): 371–93. <https://doi.org/10.1146/annurev-polisci-071112-213421>.
- Dreher, Axel, and Nathan M. Jensen. 2013. “Country or Leader? Political Change and UN General Assembly Voting.” *European Journal of Political Economy* 29: 183–96. <https://doi.org/10.1016/j.ejpoleco.2012.10.001>.
- Duque, Marina G. 2018. “Recognizing International Status: A Relational Approach.” *International Studies Quarterly* 62 (3): 577–92. <https://doi.org/10.1093/isq/sqy001>.
- Edwards, George C., and B. Dan Wood. 1999. “Who Influences Whom? The President, Congress, and the Media.” *American Political Science Review* 93 (2): 327–44. <https://doi.org/10.2307/2585399>.
- Enke, Benjamin. 2024. “The Cognitive Turn in Behavioral Economics.”
- Enke, Benjamin, and Florian Zimmermann. 2017. “Correlation Neglect in Belief Formation.” *The Review of Economic Studies*, December. <https://doi.org/10.1093/restud/rdx081>.
- Fearon, James D. 1994. “Domestic Political Audiences and the Escalation of International Dis-

- putes.” *American Political Science Review* 88 (3): 577–92. <https://doi.org/10.2307/2944796>.
- Flores-Macías, Gustavo A., and Sarah E. Kreps. 2013. “The Foreign Policy Consequences of Trade: China’s Commercial Relations with Africa and Latin America, 1992–2006.” *The Journal of Politics* 75 (2): 357–71. <https://doi.org/10.1017/S0022381613000066>.
- Folha de S.Paulo. 2016. “Alcance total dos jornais.” <https://arte.folha.uol.com.br/graficos/QcVNh/>.
- Folke, Olle, Torsten Persson, and Johanna Rickne. 2016. “The Primary Effect: Preference Votes and Political Promotions.” *American Political Science Review* 110 (3): 559–78.
- Frankel, Jeffrey, and George Saravelos. 2012. “Can Leading Indicators Assess Country Vulnerability? Evidence from the 2008–09 Global Financial Crisis.” *Journal of International Economics* 87 (2): 216–31.
- Fuchs, Andreas, and Nils-Hendrik Klann. 2013. “Paying a Visit: The Dalai Lama Effect on International Trade.” *Journal of International Economics* 91 (1): 164–77. <https://doi.org/10.1016/j.jinteco.2013.04.007>.
- Fujiwara, Thomas, and Carlos Sanz. 2020. “Rank Effects in Bargaining: Evidence from Government Formation.” *Review of Economic Studies* 87 (3): 1261–95. <https://doi.org/10.1093/restud/rdz044>.
- Garcia-Herrero, Alicia, Thibault Storella, and Pekka Weil. 2024. “China’s Influence at the United Nations: Words and Deeds.” Working Paper 19/2024. Bruegel.
- Gelman, Andrew, and Guido Imbens. 2019. “Why High-Order Polynomials Should Not Be Used in Regression Discontinuity Designs.” *Journal of Business & Economic Statistics* 37 (3): 447–56. <https://doi.org/10.1080/07350015.2017.1366909>.
- Götz, Elias. 2021. “Status Matters in World Politics.” *International Studies Review* 23 (1): 228–47. <https://doi.org/10.1093/isr/viaa046>.
- Graeber, Thomas, Christopher Roth, and Marco Sammon. 2025. “Coarse Categories in a Complex World.” University of Bonn and University of Cologne, Germany.
- Granzier, Riako, Vincent Pons, and Clemence Tricaud. 2023. “Coordination and Bandwagon Effects: How Past Rankings Shape the Behavior of Voters and Candidates.” *American Economic Journal: Applied Economics* 15 (4): 177–217. <https://doi.org/10.1257/app.20210840>.
- Gurevich, Tamara, and Peter Herman. 2018. “The Dynamic Gravity Dataset: 1948–2016.” *USITC Working Paper, 2018–02–A*.

- Haibin, Niu. 2010. "Emerging Global Partnership: Brazil and China." *Revista Brasileira de Política Internacional* 53: 183–92.
- He, Yinghua, Yuyu Zheng, and Jidong Chen. 2025. "Arms Exports and Voting Alignment at the United Nations General Assembly." *European Journal of Political Economy* 86: 102588. <https://doi.org/10.1016/j.ejpoleco.2024.102588>.
- Hirshberg, David, and Sylvia Klosin. 2024. "Synthetic Differences-in-Differences with Covariates." https://klosins.github.io/Hirshberg_Klosin_SDIDC.pdf.
- Imai, Kosuke, and In Song Kim. 2019. "When Should We Use Unit Fixed Effects Regression Models for Causal Inference with Longitudinal Data?" *American Journal of Political Science* 63 (2): 467–90. <https://doi.org/10.1111/ajps.12417>.
- . 2021. "On the Use of Two-Way Fixed Effects Regression Models for Causal Inference with Panel Data." *Political Analysis* 29 (3): 405–15.
- Imai, Kosuke, In Song Kim, and Erik H. Wang. 2023. "Matching Methods for Causal Inference with Time-Series Cross-Sectional Data." *American Journal of Political Science* 67 (3): 587–605. <https://doi.org/10.1111/ajps.12685>.
- Jabarian, Brian, and Elia Sartori. 2024. "Critical Thinking and Storytelling Contexts." <https://doi.org/10.2139/ssrn.4991780>.
- Kastner, Scott L., and Margaret M. Pearson. 2021. "Exploring the Parameters of China's Economic Influence." *Studies in Comparative International Development* 56 (1): 18–44. <https://doi.org/10.1007/s12116-021-09318-9>.
- Lal, Apoorva, Mackenzie Lockhart, Yiqing Xu, and Ziwen Zu. 2024. "How Much Should We Trust Instrumental Variable Estimates in Political Science? Practical Advice Based on 67 Replicated Studies." *Political Analysis* 32 (4): 521–40.
- Lektzian, David, and Glen Biglaiser. 2023. "US Economic Statecraft and Voting at the United Nations General Assembly." *International Interactions* 49 (1): 115–43. <https://doi.org/10.1080/03050629.2022.2156928>.
- Liu, Licheng, Ye Wang, and Yiqing Xu. 2024. "A Practical Guide to Counterfactual Estimators for Causal Inference with Time-Series Cross-Sectional Data." *American Journal of Political Science* 68 (1): 160–76.
- MacDonald, Paul K., and Joseph M. Parent. 2021. "The Status of Status in World Politics." *World*

- Politics* 73 (2): 358–91. <https://doi.org/10.1017/s0043887120000301>.
- MacKinnon, James G., and Matthew D. Webb. 2018. “The Wild Bootstrap for Few (Treated) Clusters.” *The Econometrics Journal* 21 (2): 114–35. <https://doi.org/10.1111/ectj.12107>.
- Mellon, Jonathan. 2021. “Rain, Rain, Go Away: 194 Potential Exclusion-Restriction Violations for Studies Using Weather as an Instrumental Variable.” *American Journal of Political Science*.
- Mercer, Jonathan. 2017. “The Illusion of International Prestige.” *International Security* 41 (4): 133–68. https://doi.org/10.1162/isec_a_00276.
- Miller, Jennifer L., Jacob Cramer, Thomas J. Volgy, Paul Bezerra, Megan Hauser, and Christina Sciabarra. 2015. “Norms, Behavioral Compliance, and Status Attribution in International Politics.” *International Interactions* 41 (5): 779–804. <https://doi.org/10.1080/03050629.2015.1037709>.
- Mintz, Alex. 2004. “How Do Leaders Make Decisions? A Poliheuristic Perspective.” *Journal of Conflict Resolution* 48 (1): 3–13.
- . 2005. “Applied Decision Analysis: Utilizing Poliheuristic Theory to Explain and Predict Foreign Policy and National Security Decisions.” *International Studies Perspectives* 6 (1): 94–98. <https://doi.org/10.1111/j.1528-3577.2005.00195.x>.
- Mintz, Alex, and Nehemia Geva. 1997. “The Poliheuristic Theory of Foreign Policy Decision-making.” In *Decisionmaking on War and Peace: The Cognitive–Rational Debate*, edited by Nehemia Geva and Alex Mintz, 81–101. Boulder, CO: Lynne Rienner.
- Morgan, Pippa. 2019. “Can China’s Economic Statecraft Win Soft Power in Africa? Unpacking Trade, Investment and Aid.” *Journal of Chinese Political Science* 24 (3): 387–409. <https://doi.org/10.1007/s11366-018-09592-w>.
- Mourón, Fernando, and Francisco Urdinez. 2014. “A Comparative Analysis of Brazil’s Foreign Policy Drivers Towards the USA: Comment on Amorim Neto (2011).” *Brazilian Political Science Review* 8 (2): 94–115. <https://doi.org/10.1590/1981-38212014000100013>.
- Müller, Karsten, Chenzi Xu, Mohamed Lehib, and Ziliang Chen. 2025. “The Global Macro Database: A New International Macroeconomic Dataset.” Working Paper Series. National Bureau of Economic Research. <https://doi.org/10.3386/w33714>.
- Neto, Octavio Amorim. 2011. *De Dutra a Lula: A Condução e Os Determinantes Da Política Externa Brasileira*. Elsevier Brasil.

- Neto, Octavio Amorim, and Andrés Malamud. 2015. "What Determines Foreign Policy in Latin America? Systemic Versus Domestic Factors in Argentina, Brazil, and Mexico, 1946–2008." *Latin American Politics and Society* 57 (4): 1–27.
- Parlar Dal, Emel. 2019. "Status Competition and Rising Powers in Global Governance: An Introduction." *Contemporary Politics* 25 (5): 499–511. <https://doi.org/10.1080/13569775.2019.1627767>.
- Potrafke, Niklas. 2009. "Does Government Ideology Influence Political Alignment with the US? An Empirical Analysis of Voting in the UN General Assembly." *The Review of International Organizations* 4 (3): 245–68.
- Presidência da República Federativa do Brasil. 2009. "Discurso Do Presidente Da República Luiz Inácio Lula Da Silva Por Ocasão Do Seminário Brasil-China: Novas Oportunidades Para a Parceria Estratégica — Pequim, 19 de Maio de 2009." <https://www.gov.br/mre/pt-br/centrais-de-conteudo/publicacoes/discursos-artigos-e-entrevistas/presidente-da-republica/presidente-da-republica-federativa-do-brasil-discursos/luiz-inacio-lula-da-silva-2003-2007/discurso-do-presidente-da-republica-luiz-inacio-lula-da-silva-por-ocasio-do-seminario-brasil-china-novas-oportunidades-para-a-parceria-estrategica-pequim-19-de-maio-de-2009>.
- Reilly, James. 2013. "China's Economic Statecraft: Turning Wealth into Power." Analysis. Sydney: Lowy Institute for International Policy. https://www.lowyinstitute.org/sites/default/files/reilly_chinas_economic_statecraft_web.pdf.
- Renshon, Jonathan. 2016. "Status Deficits and War." *International Organization* 70 (3): 513–50. <https://doi.org/10.1017/s0020818316000163>.
- Renshon, Jonathan, Allan Dafoe, and Paul Huth. 2018. "Leader Influence and Reputation Formation in World Politics." *American Journal of Political Science* 62 (2): 325–39. <https://doi.org/10.1111/ajps.12335>.
- Rodrigues, Azelma. 2009. "China desbanca os EUA como maior parceiro comercial do Brasil." <https://oglobo.globo.com/economia/china-desbanca-os-eua-como-maior-parceiro-comercial-do-brasil-3170484>; O Globo.
- Røren, Pål. 2024. "Status Orders: Toward a Local Understanding of Status Dynamics in World Politics." *International Studies Review* 26 (4). <https://doi.org/10.1093/isr/viae044>.
- Schenoni, Luis L., and Diego Leiva. 2021. "Dual Hegemony: Brazil Between the United States and

- China.” In *Hegemonic Transition*, edited by Florian Böller and Welf Werner, 233–55. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-74505-9_12.
- Sheffer, Lior, Peter John Loewen, Stuart Soroka, Stefaan Walgrave, and Tamir Sheafer. 2018. “Nonrepresentative Representatives: An Experimental Study of the Decision Making of Elected Politicians.” *American Political Science Review* 112 (2): 302–21. <https://doi.org/10.1017/S0003055417000569>.
- Signorino, Curtis S, and Jeffrey M Ritter. 1999. “Tau-b or Not Tau-b: Measuring the Similarity of Foreign Policy Positions.” *International Studies Quarterly* 43 (1): 115–44.
- Struver, G. 2014. “‘Bereft of Friends’? China’s Rise and Search for Political Partners in South America.” *The Chinese Journal of International Politics* 7 (1): 117–51. <https://doi.org/10.1093/cjip/pot018>.
- Strüver, Georg. 2016. “What Friends Are Made of: Bilateral Linkages and Domestic Drivers of Foreign Policy Alignment with China.” *Foreign Policy Analysis* 12 (2): 170–91.
- Tanner, Murray Scot. 2007. *Chinese Economic Coercion Against Taiwan: A Tricky Weapon to Use*. Rand Corporation.
- Tomz, Michael. 2007. “Domestic Audience Costs in International Relations: An Experimental Approach.” *International Organization* 61 (4): 821–40. <https://doi.org/10.1017/S0020818307070282>.
- Violos, John, Konstantinos Tserpes, Iraklis Varlamis, and Theodora Varvarigou. 2018. “Text Classification Using the n-Gram Graph Representation Model over High Frequency Data Streams.” *Frontiers in Applied Mathematics and Statistics* 4: 41.
- Ward, Steven Michael. 2017. “Lost in Translation: Social Identity Theory and the Study of Status in World Politics.” *International Studies Quarterly* 61 (4): 821–34. <https://doi.org/10.1093/isq/sqx042>.
- Wheeler, Nicholas J. 2021. “Arms Imports and Shifts in Voting in the United Nations General Assembly.” *European Journal of International Relations* 27 (4): 1178–1202. <https://doi.org/10.1177/13540661211032984>.
- Yan, Jiaqiang, and Yonghong Zhou. 2021. “Economic Return to Political Support: Evidence from Voting on the Representation of China in the United Nations.” *Journal of Asian Economics* 75 (August): 101325. <https://doi.org/10.1016/j.asieco.2021.101325>.

9 Appendix

We present more details about the performance and checks on the fitted model and how we used ChatGPT to perform the NLP analysis.

9.1 SDiD

9.1.1 Comparison of DiD, SCM, and SDiD estimating equations

To gain intuition for the SDiD estimator, it is useful to compare it with standard DiD and SCM. A DiD model estimates:

$$(\hat{\tau}_{ATT}^{DiD}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\tau_{ATT}, \mu, \alpha, \beta} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \mu - \alpha_i - \beta_t - D_{it}\tau_{ATT})^2$$

While SDiD assigns weights to the control group that are more similar to those of the treated unit and in time periods comparable to the pre-treatment period, classic DiD does not use any weights at all.

The SCM model can be written as follows:

$$(\hat{\tau}_{ATT}^{SCM}, \hat{\mu}, \hat{\beta}) = \arg \min_{\tau_{ATT}, \mu, \beta} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \mu - \beta_t - D_{it}\tau_{ATT})^2 \hat{w}^{SCM}$$

The equations show there is no unit fixed effect and no time weight in SCM. As a result, SCM cannot use unit-specific unobserved invariant characteristics to approximate the counterfactual outcome, and must consider the entire period when estimating the causal effect. SDiD combines unit weights (like SCM) with unit fixed effects and time weights (like DiD), addressing the limitations of both. The basic SDiD estimating equation is:

$$(\hat{\tau}_{ATT}^{SDiD}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\tau_{ATT}, \mu, \beta, \alpha} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \mu - \alpha_i - \beta_t - D_{it}\tau_{ATT})^2 \hat{w}^{SDiD} \hat{\lambda}_t^{SDiD}$$

Note that SDiD retains unit fixed effects α_i (like DiD) while also assigning unit weights $\hat{w}^{SDiD}$ (like SCM) and time weights $\hat{\lambda}_t^{SDiD}$ (unique to SDiD).

9.1.2 SDiD weights and donor pool

Below we present the weights for the top 10 controls used in the main model.

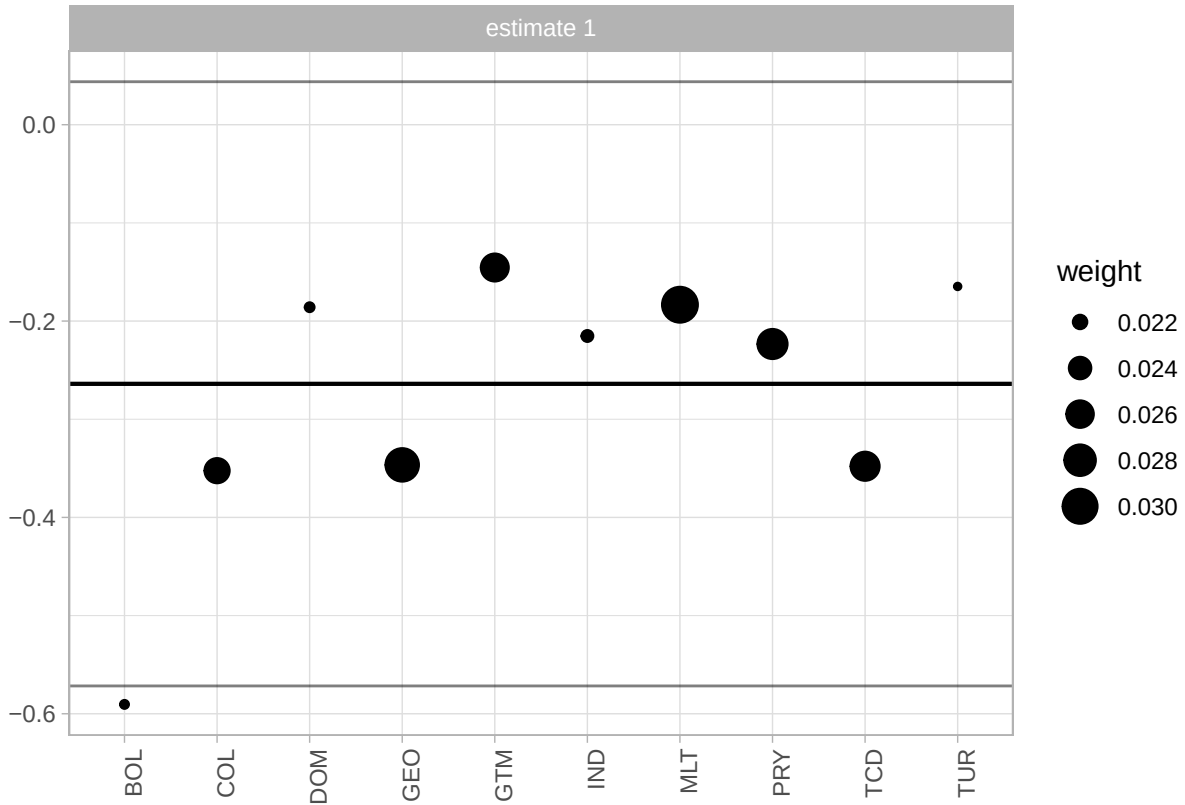


Figure 13: Weights of countries contributing to the synthetic control in the Brazilian SDiD model. Note: Unit = synthetic-control unit weights (unitless, sum to 1). Time window = weights are estimated from the pre-treatment period ending in 2008. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

We also fit a model for a donor pool of Latin American states (excluding Caribbean countries), which is more similar to Brazil than using countries worldwide. One reason to prefer a pool of Latin American countries is that a smaller donor pool helps avoid overfitting. When comparing results, the estimated causal effect is quite similar (a bit larger in absolute terms), which lends credibility to the main estimation. However, the small number of controls introduces noise and increases the uncertainty of the model. It is known that the usage of the placebo method to compute standard errors inflates the uncertainty of the estimation. For this reason, we prefer the model with the larger donor pool. Nonetheless, it is reassuring that with a donor pool composed of Latin American

countries, which are arguably more similar to Brazil, the estimate is quite similar. Below is the plot of the model fitted for Latin America.

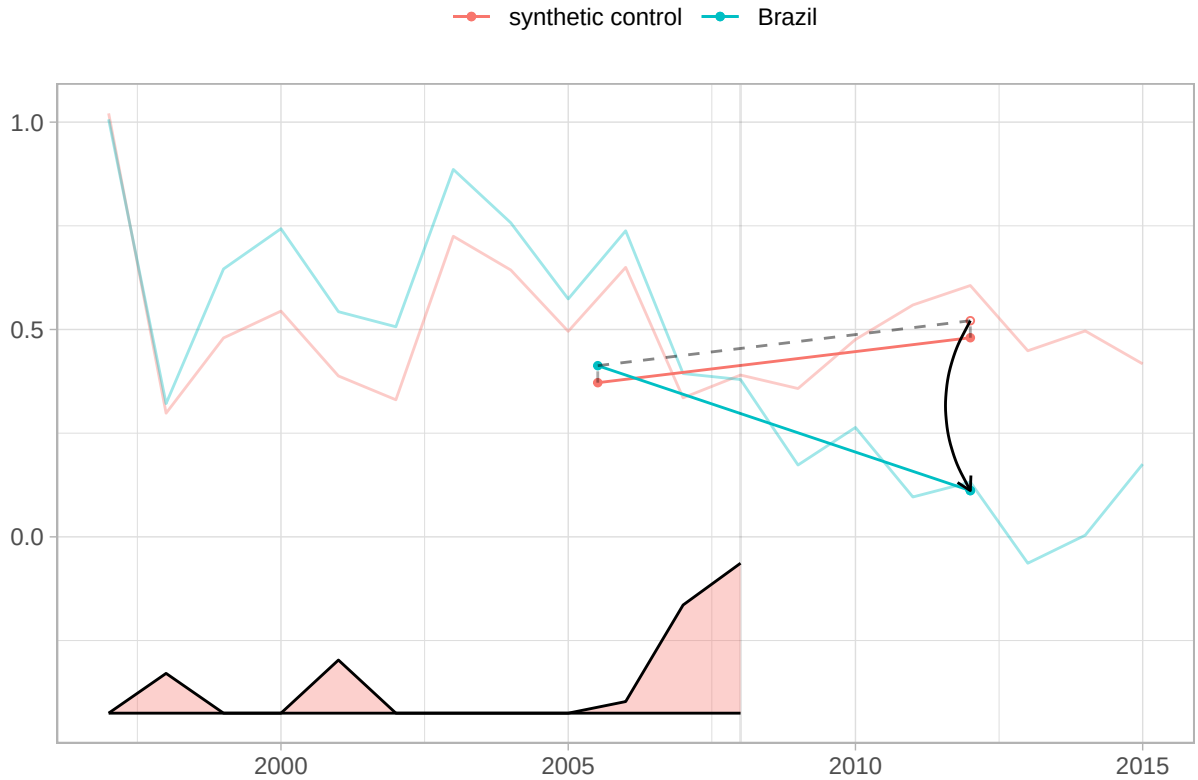


Figure 14: (#fig:plot latam)SDiD fit with a Latin American donor pool. Note: Unit = absolute UNGA ideal-point distance to China (points). Standard error = placebo-based (reported in text; not plotted here). Time window = pre-treatment and post-treatment years shown in the plot (treatment in 2009). Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil’s top export destination displacing the USA, and 0 before 2009.

The estimate is -0.41 with a standard error of 0.2. In any case, it is interesting to examine the countries with the largest weights. Argentina, Guatemala, Paraguay, and Uruguay – all but Guatemala are founding members of Mercosur. They contribute the most to synthetic control.

9.2 Cross-Country Robustness: Absorbing-Treatment Estimator (C&S)

As a robustness check, we also report results from the Callaway and Sant’Anna (2021) absorbing-treatment estimator, restricted to countries where China’s displacement of the USA was never reversed (7 countries). This estimator assumes that treatment is permanent once assigned – an as-

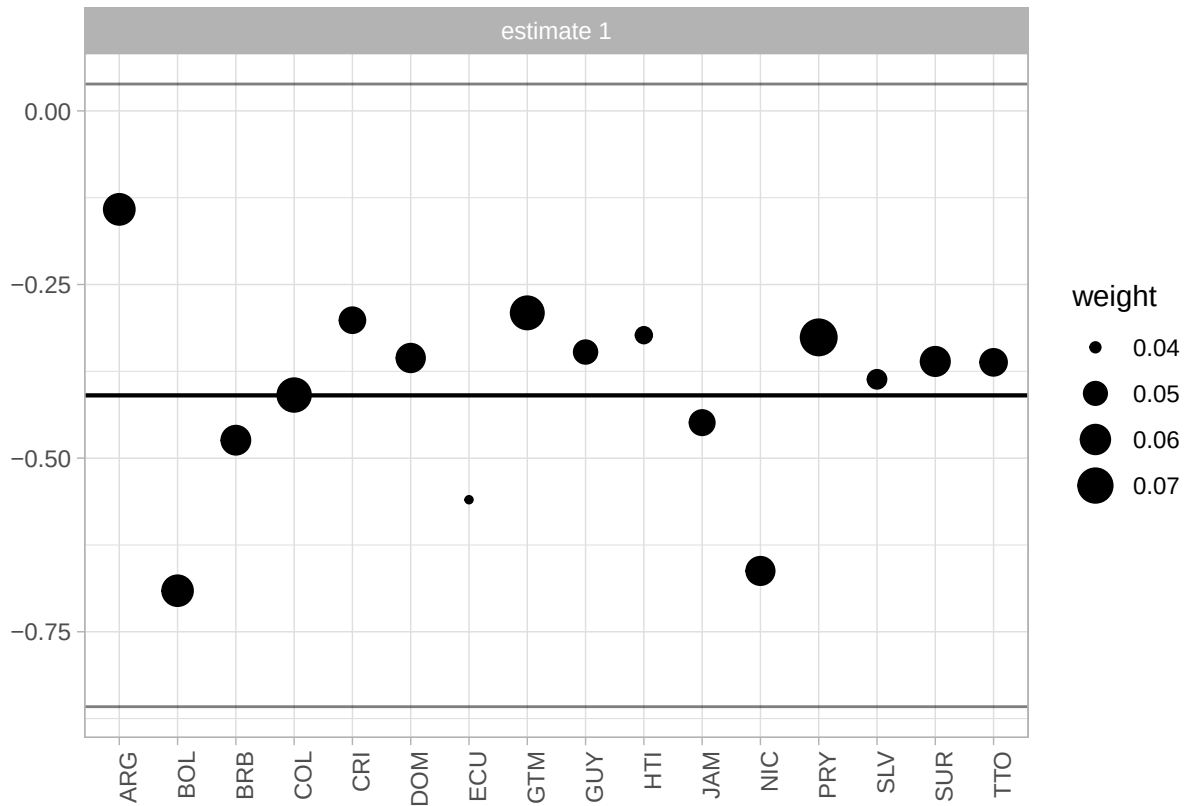


Figure 15: (#fig:plot weight latam) Synthetic-control donor weights for the Latin American SDiD specification. Note: Unit = synthetic-control unit weights (unitless, sum to 1). Time window = weights are estimated from the pre-treatment period ending in 2008. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

sumption violated for 52 of the 59 treated countries but appropriate for the absorbing subset. The absorbing-only C&S ATT (-0.117, SE = 0.045, p = 0.009) is directionally consistent with the fect IFE result, confirming that the finding is not an artifact of the switching-treatment framework.

With only 7 treated countries in the absorbing subset, asymptotic cluster-robust inference may be liberal. We address this concern with two finite-sample procedures applied to the full C&S sample.

Wild cluster bootstrap. We stack the staggered treatments into a single DiD specification and apply the wild cluster bootstrap with Webb weights (MacKinnon and Webb 2018), using 9,999 replications. The stacked DiD coefficient is -0.160 (cluster SE = 0.046) with a bootstrap p-value of 0.018 and 95% bootstrap CI [-0.275, -0.044], based on 67 clusters. The result remains statistically significant under this more conservative inference procedure.

Fisher randomization test. To provide exact inference that does not rely on large-sample asymptotics, we randomly reassign treatment status across countries and re-estimate the overall ATT under each permutation. The observed ATT (-0.117) is more extreme than 96.7% of the 1000 valid permutation estimates (Fisher p = 0.033).

9.3 Cross-Country Diagnostic: No-Pretrend Equivalence Tests

Following Liu, Wang, and Xu (2024) and Chiu et al. (2025), we report equivalence tests for the no-pretrend assumption. Unlike the conventional placebo test (which tests H_0 : no pre-trend), the equivalence test reverses the null to H_0 : pre-trends are meaningfully large, providing positive evidence for the absence of pre-trends when rejected. The red dashed lines mark the equivalence range ($0.36\hat{\delta}_s$ or the user-specified threshold), while the grey dashed lines mark the minimum range.

9.4 ChatGPT Classification

Below is the coded instruction to the ChatGPT API to classify the subjects of the headlines.

```
system_block <- paste(
  'You are a Brazilian expert in International Relations and the political',
  'economy of China-Brazil relations. Classify the subject of EACH headline',
  'when I say EACH, I mean all of them, even if repetitive or similar. Do classify **all
```

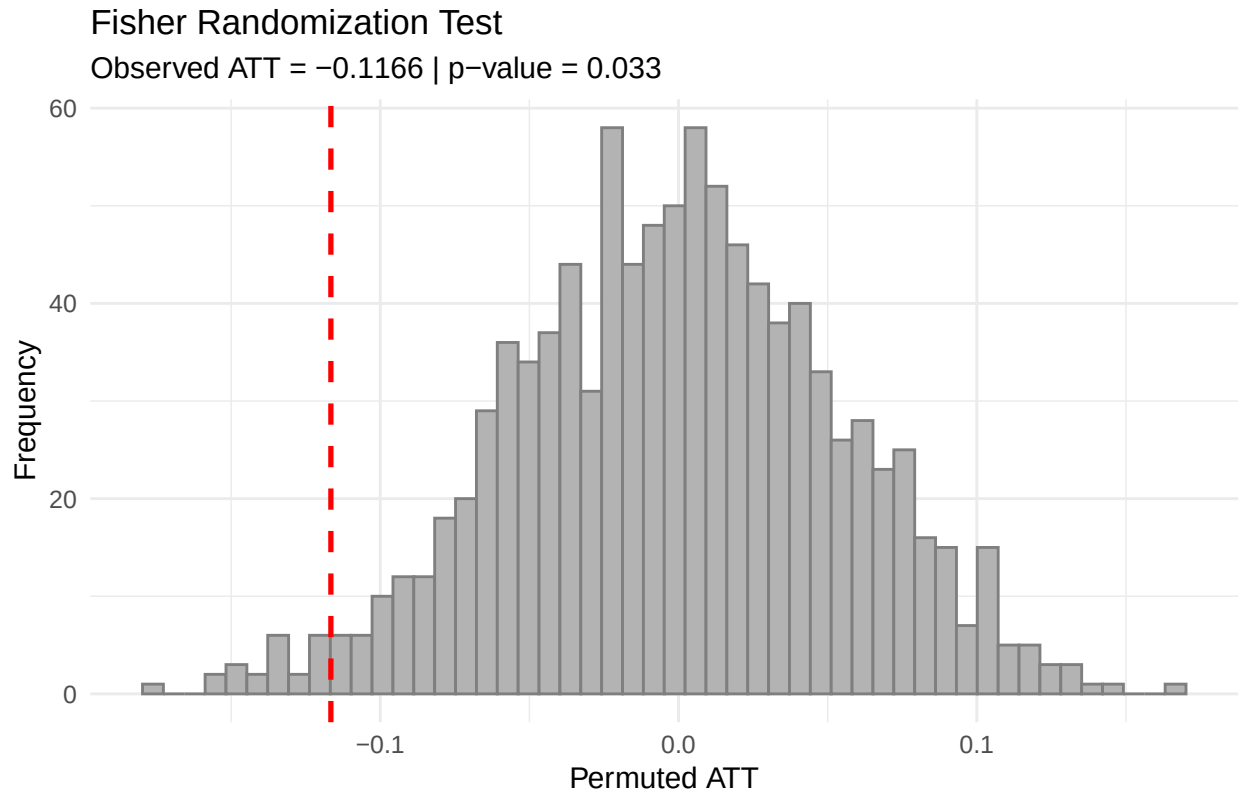


Figure 16: Fisher randomization distribution with the observed ATT marked by the dashed red line. Note: Unit = ATT in absolute UNGA ideal-point distance to China (points). Standard error = permutation-based exact inference. Time window = 1990-2022 balanced panel used for permutations. Treatment definition = Treatment indicator equals 1 in any year China is currently the top export destination; turns off if USA regains the position (switching treatment).

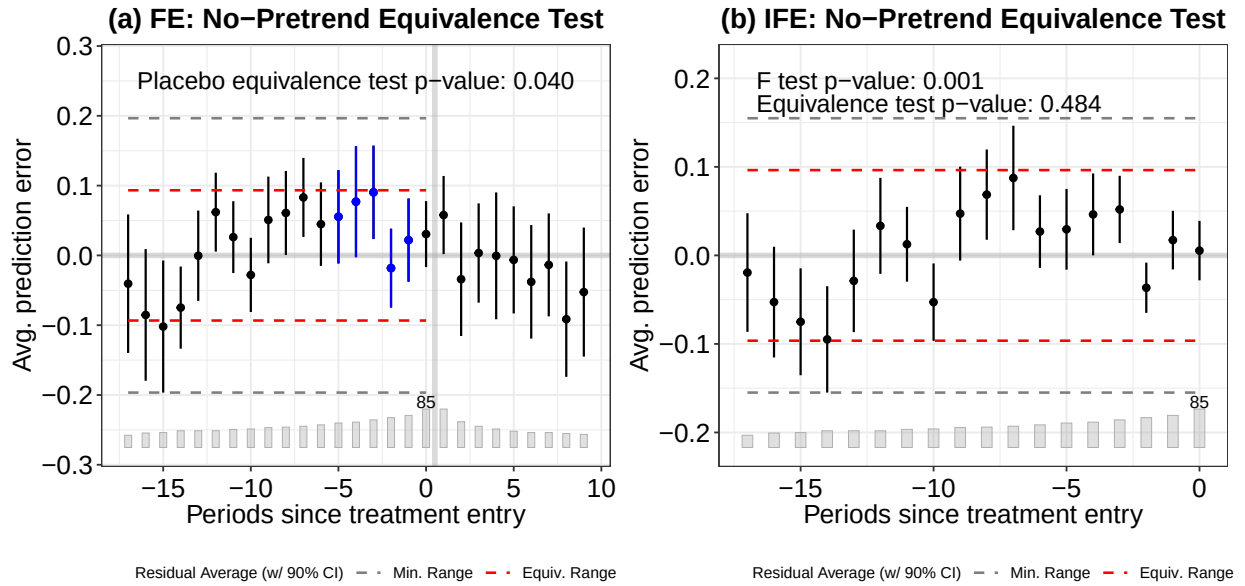


Figure 17: Equivalence tests for the no-pretrend assumption. (a) FE model: placebo equivalence test rejects the null of large pre-trends (TOST $p = 0.040$). (b) IFE model: the F-test rejects the null of zero pre-trends ($p = 0.001$), though the equivalence test does not reject (TOST $p = 0.484$), reflecting the limited power of the small cross-country sample. Note: Standard error = bootstrap (500 replications). Time window = 1990-2023 panel. Treatment definition = Treatment indicator equals 1 in any year China is currently the top export destination; turns off if USA regains the position (switching treatment).

```

'using one label from this list exactly as written:',
'diplomacy', 'chinese economy', 'china-brasil relations',',
'disaster and accidents', 'chinese politics and policy',',
'sports, science, culture, other', 'human rights',',
'china-brazil trade', 'health'.\n\n',
# 13 static few-shot examples - leave them unchanged!
'Examples:\n',
'1. Presidente chinês visita o Brasil e participa de reunião com FHC -> china-brasil r
'2. Ministro chinês de Defesa faz visita oficial ao Brasil -> china-brasil relations\n
'3. Para governo, novo câmbio chinês só beneficia Brasil no médio prazo -> china-brazi
'4. China ultrapassa EUA como maior parceiro comercial do Brasil -> china-brazil trade
'5. China vai controlar preços de alimentos e combustíveis -> chinese economy\n',
'6. Vendas de automóveis na China desaceleram em junho -> chinese economy\n',
'7. China quer devolução das relíquias Yin, patrimônio da Unesco -> sports, science, c
'8. Dinossauro encontrado na China esclarece origem das aves -> sports, science, cultu
'9. OMS confirma mais de 4.600 casos de gripe suína; China registra 2º caso -> health\
'10. Marinha chinesa busca "cansar" barcos japoneses em águas disputadas -> diplomacy\
'11. Acidente após casamento deixa 16 mortos na China -> disaster and accidents\n',
'12. Jornal chinês sai das bancas por foto da praça Tiananmen -> human rights\n',
'13. China é parceira estratégica contra crise econômica, diz UE -> chinese economy',
'return as in the examples, a string with the headline and the classification'
)

```

It is important to highlight that the process was not streamlined. We faced several problems, such as ChatGPT refusing to classify subjects (saying they were repetitive and too similar), changing the original headline, which made matching back to the original dataset difficult, and other similar problems. As a result, we had to rerun the instructions a few times on a sample and tweak it with some prompt engineering until arriving at that final version. As a result, this is the only part of the paper not fully reproducible, in the sense that another request for classification, with the same instructions, would yield a few different categorizations and a few different original headlines

Table 5: Validation of ChatGPT classification against manual coding (N = 100). Overall accuracy: 88.0 percent.

Category	N	Correct	Accuracy (%)
china-brazil trade	6	6	100.0
disaster and accidents	13	13	100.0
health	6	6	100.0
chinese economy	25	24	96.0
human rights	8	7	87.5
diplomacy	12	10	83.3
sports, science, culture, other	17	14	82.4
china-brazil relations	5	4	80.0
chinese politics and policy	8	4	50.0
Overall	100	88	88.0

changed, along with some refusals to classify texts.

Below we provide a random sample of the headlines and the classification provided by ChatGPT.

9.4.1 Validation of ChatGPT Classification

To assess the reliability of the automated classification, one of the authors independently coded a stratified random sample of 100 headlines (with a minimum of 5 per category). The table below reports the agreement rate between the ChatGPT labels and the manual coding.

Note: Unit = headline-level counts and accuracy (percent) by category. Time window = stratified validation sample from the 2000-2014 headline corpus.

The overall accuracy is 88%. The lowest agreement is in the “chinese politics and policy” category (50%), where several headlines were manually reclassified as “diplomacy”. The categories most relevant to our analysis — “china-brazil trade” and “disaster and accidents” — achieved 100% agreement.

9.5 Cross-Country: Treated Countries and Treatment-Switching Diagnostic

The table below describes the 59 treated countries where China displaced the United States as the top trade partner. For each country, we report the treatment year, whether the treatment is absorbing (China remains #1) or switching (the USA regains the top position), the number of consecutive years

TABLE 3 — Examples of Classified Headlines	
Random sample of 20 headlines with ChatGPT-assigned topic	
Headline	Date
china-brazil relations	
Lula vê apoio chinês a ambição do Brasil na ONU	2004-11-13
Para executivo chinês, modelo brasileiro de infraestrutura é "piada"	2010-09-17
china-brazil trade	
Diretor do Ciesp diz que câmbio pressiona mais que salvaguardas contra China	2005-05-24
Acordo entre Brasil e China sobre soja ainda deixa dúvidas	2004-06-23
chinese economy	
China tem crescimento de 8,2% na economia mesmo com a Sars	2003-07-16
Movimento do principal porto do mundo, na China, é o menor em 7 anos	2008-12-16
chinese politics and policy	
Presidente chinês promete "limpar" a internet	2007-01-24
China quer evitar estímulo a "rebeldes"	2001-04-26
diplomacy	
Rússia e China aumentam pressão sobre Iraque	2002-09-14
China diz se preocupar com o Irã, mas ainda rejeita sanções	2010-03-16
disaster and accidents	
Explosões matam 7 em prédio residencial na China	2002-02-05
Chuva de granizo mata 13 pessoas no sul da China	2007-04-02
health	
China registra três novos casos de Sars	2003-06-03
OMS confirma caso suspeito de gripe aviária em humanos na China	2005-11-06
human rights	
Governo chinês prende policiais que deixaram mulher ter nove filhos	2006-05-08
China intensifica repressão contra o Tibet, diz dalai-lama	2001-04-02
sports, science, culture, other	
China lança primeira sonda espacial destinada a explorar a Lua	2013-11-30
Seleção feminina desembarca na China para o Mundial de basquete	2002-09-11

Figure 18: Random sample of China-related Folha de Sao Paulo headlines and ChatGPT classifications. Note: Unit = headline text and category labels. Time window = sample drawn from the 2000-2014 corpus.

the US held the top position before treatment (USA streak), and the mean UNGA voting distance to China before and after treatment.

Table 6: Descriptive statistics of treated countries where China displaced the USA as top trade partner.

Country	ISO3c	Treatment year	Absorbing	Switches	USA streak	Press free.	Mean dist. (pre)	Mean dist. (post)	Diff
South Korea	KOR	2003	No	2	15	0.952	1.536	1.438	-0.099
Equatorial Guinea	GNQ	2006	No	7	4	0.165	0.674	0.341	-0.333
Angola	AGO	2007	No	2	16	0.337	0.513	0.247	-0.266
Chile	CHL	2008	No	2	10	0.942	0.800	0.457	-0.343
Brazil	BRA	2009	No	2	20	0.930	0.733	0.408	-0.325
Thailand	THA	2009	No	2	20	0.609	0.461	0.258	-0.203
Japan	JPN	2010	No	6	22	0.925	1.736	1.326	-0.410
Peru	PER	2012	No	4	21	0.914	0.758	0.418	-0.340
Saudi Arabia	SAU	2015	No	2	3	0.120	0.386	0.220	-0.166
Vietnam	VNM	2018	No	5	16	0.150	0.548	0.332	-0.216

Note:

Absorbing = China remains the top trade partner through the end of the sample. Switching = USA regains the top position. USA streak = consecutive years the US held #1 export-destination position before treatment. Press free. = V-Dem Freedom of Expression index (pre-treatment 5-year mean). Mean dist. = absolute distance to China in UNGA ideal points. Diff = post minus pre.

The dynamic event study below displays the estimated ATT at each period relative to treatment across C&S specifications (absorbing-only). These are provided for comparison with the main fact IFE results reported in the body of the paper.

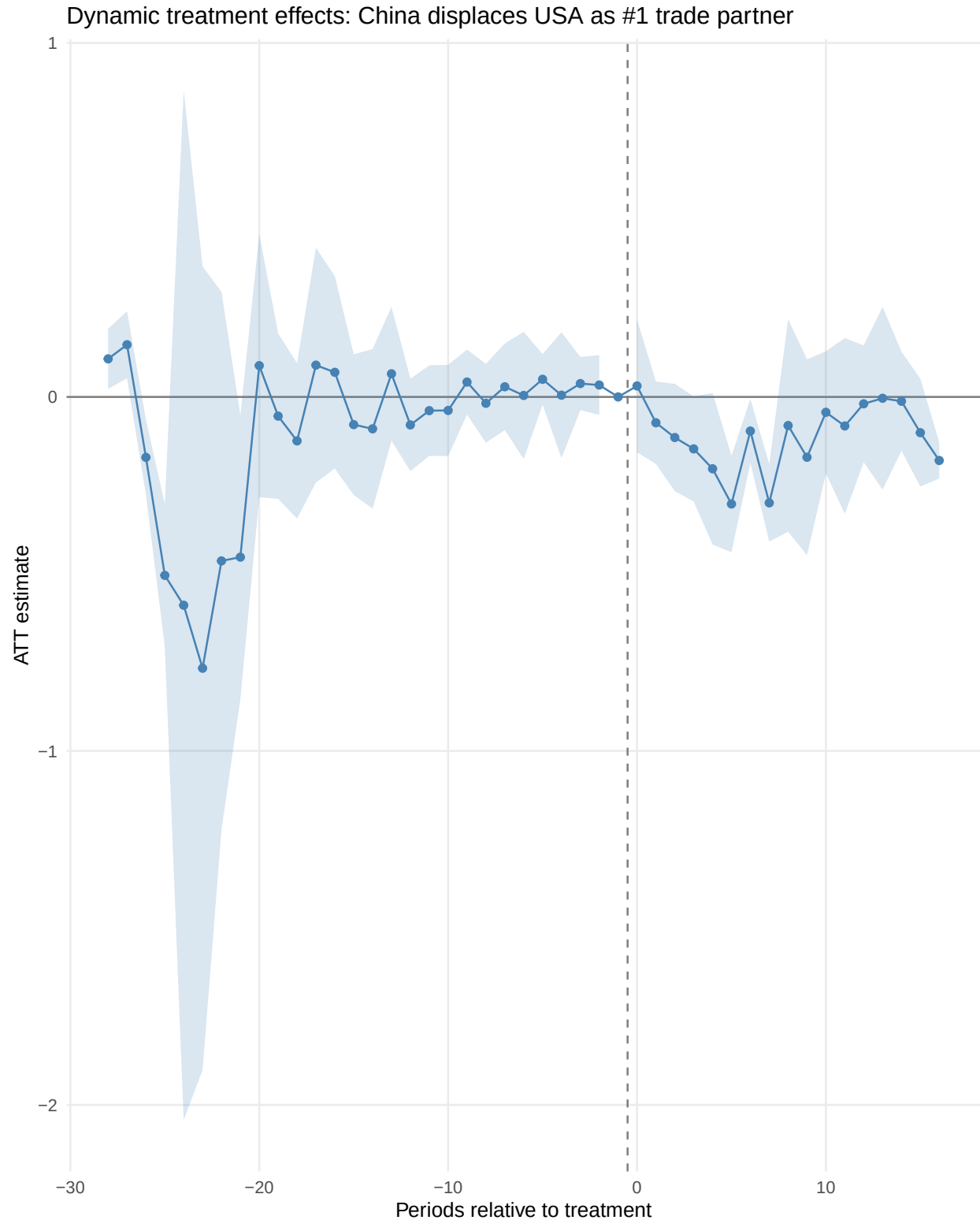


Figure 19: Dynamic treatment effects across C&S specifications (absorbing treatment). Note: Unit = ATT in absolute UNGA ideal-point distance to China (points). Standard error = analytical (95 percent confidence intervals shown). Cluster = country level. Time window = 1990-2022 baseline panel and 1995-2022 restricted panel, depending on specification. Treatment definition = absorbing treatment: equals 1 from first year China becomes top partner, assumes no reversal.